Exploiting GANs Against IDSs: A Systematic Review, Meta-Analysis, and Case Study Evaluation

Yasmeen Alslman, Mouhammd Alkasassbeh, and Mohammad J. Abdel-Rahman, *Senior Member, IEEE*

***Abstract*—With the rapid improvement of generative adversarial networks (GANs) in the landscape of data augmentation and generation, researchers have started to explore the use of GANs in other fields, such as adversarial attack generation. Using GANs has improved the field of adversarial attacks, particularly against intrusion detection systems (IDSs). GAN-based adversarial attacks are considered sophisticated attacks that can significantly compromise the security of IDSs. Addressing GAN-based adversarial challenges can ensure the effectiveness and severity of such attacks, which, in turn, lead to the development of strong defense strategies. Few surveys have covered the effects and challenges of adversarial attacks against IDSs, with a focus on gradient-based adversarial attacks. Recently, many research articles have examined GAN-based adversarial attacks. This article provides a systematic and comprehensive review of recent research on GAN-based adversarial attacks targeting IDSs, complemented by a meta-analysis guided by critical review questions. Through case studies, we empirically demonstrate the vulnerabilities of binary and multiclass IDS models under GAN-based attacks, examining the influence of feature selection. We are the first to employ advanced GAN variants to craft novel adversarial attacks. Using Fréchet inception distance and Hellinger distance, we assess IDS performance and the realism of generated samples, revealing key system weaknesses. The article concludes by summarizing the current state of knowledge, outlining major challenges, proposing future research directions, and identifying the most severe GAN architecture and the one that generates the most realistic adversarial samples.**

***Impact Statement*—While gradient-based adversarial attacks on IDSs have been comprehensively studied and analyzed, GAN-based adversarial attacks are not adequately covered in existing reviews. This work enables researchers to understand GAN-based adversarial attacks against IDSs by providing a brief background on such attacks and their current defense mechanisms. Literature on GAN-based adversarial attacks has been categorized to help researchers comprehend the current progress in this field. With our meta-analysis, researchers can gain insights into the commonly used approaches in GAN-based adversarial attacks and identify gaps in such attacks regarding the used GAN architectures, the targeted models, and other aspects. The representative use cases examined in this work enlighten researchers on the effects of GAN-based attacks on IDSs. They also inform researchers of the shortcomings that must be addressed when attacking and defending against such attacks.**

***Index Terms*—Adversarial attacks, adversarial machine learning, deep learning, generative adversarial attacks, intrusion detection systems (IDSs), machine learning.**

Received 25 May 2025; revised 23 September 2025, 18 November 2025, and 26 January 2026; accepted 16 March 2026. Date of publication 23 March 2026; date of current version 31 August 2026. This article was recommended for publication by Associate Editor C. L. Phillip Chen upon evaluation of the reviewers' comments. *(Corresponding author: Mohammad J. Abdel-Rahman.)*

Yasmeen Alslman and Mouhammd Alkasassbeh are with the Department of Computer Science, Princess Sumaya University for Technology, Amman 11941, Jordan.

Mohammad J. Abdel-Rahman is with the Department of Data Science, Princess Sumaya University for Technology, Amman 11941, Jordan, and also with the Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA 24061 USA (e-mail: m.AbdelRahman@psut.edu.jo).

Digital Object Identifier 10.1109/TAI.2026.3677234

I. Introduction

Generative adversarial networks (GANs) were introduced by Goodfellow et al. [1], consisting of two main networks, a generator and a discriminator, which were able to produce highly realistic synthetic data. Both these networks (i.e., generator and discriminator) are trained simultaneously, where the generator aims to create synthetic data indistinguishable from real data. On the other hand, the discriminator attempts to differentiate between real and generated data. GANs have been utilized in various domains, such as image generation [2], data augmentation [3], unsupervised learning [4], and cybersecurity [5], [6].

In cybersecurity, intrusion detection systems (IDSs) play a significant role in preventing cyber threats. With the evolving nature of these threats, integrating GANs into IDSs has emerged as a promising approach to enhance the detection capabilities of IDSs, as demonstrated in [5], [6]. In IDSs, GANs can be used in two different ways. First, as a model to construct balanced datasets by generating samples from minority classes (i.e., attacks), such as in [7], [8]. Second, as a classification model for attack detection, as proposed in [9].

GANs can enhance IDSs by providing diverse training data; however, they can also be leveraged to craft sophisticated adversarial attacks that challenge the robustness of IDSs. GANs can be used to generate adversarial attacks, where the generator learns how to introduce noise into abnormal traffic to mislead IDSs into classifying such traffic as normal traffic. At the same time, the discriminator learns to mimic the IDS. As illustrated in Fig. 1, adversarial attacks can be classified based on the attacker's knowledge, the attacker's goal, and the method by which attacks are created. The main focus of our article is on black-box, targeted, and GAN-based adversarial attacks against IDSs.

The use of GANs in cybersecurity is well-established; considerable work has summarized the applications of GANs in cybersecurity [10], [11]. However, these works often overlook the impact of GAN-based adversarial attacks, which can affect

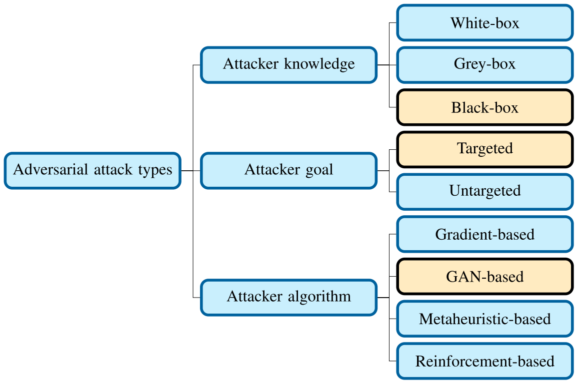

Fig. 1. Adversarial attack types.

the reliability of security systems. Numerous studies address the consequences of adversarial attacks on machine learning (ML) and deep learning (DL) models [12], [13], [14], [15]. Nevertheless, these studies have primarily focused on gradient-based attacks. Unlike gradient-based, GAN-based attacks leverage the capability of producing highly sophisticated adversarial examples. Despite the growing interest in this area, semantic challenges posed by GAN-based adversarial attacks against systems such as IDSs have yet to be thoroughly explored.

Motivated by the evolving threats shaped by GAN-based adversarial attacks, we systematically and comprehensively review the literature related to GAN-based adversarial attacks on IDSs. We identify the existing types of GANs used for generating adversarial attacks, the types of target models, the datasets used, and the metrics used to evaluate the robustness of the threat models. By identifying and classifying existing research on GAN-based adversarial attacks, analyzing the common methods used to build IDSs and generate adversarial attacks, reviewing the defense strategies proposed in the literature to counter such attacks, and assessing the effectiveness of these attacks, our review develops valuable insights and future directions for researchers in this area.

Main Contributions—Our main contributions in this article can be summarized as follows.

1) Conduct a systematic review, following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) protocol [16]. Inclusion and exclusion criteria are applied to 274 papers collected from four main digital libraries. The quality of the resulting 42 papers is evaluated based on the diversity of the datasets and IDSs used, the existence of a comparative evaluation with baselines, and the novelty of the examined GAN-based adversarial attacks. As a result, six papers are excluded.
2) Review the 36 identified papers comprehensively. Furthermore, we present a detailed overview covering all stages of generating GAN-based adversarial attacks, from building the targeted IDS to assessing and defending it against GAN-based adversarial attacks.
3) Conduct a meta-analysis based on critical review questions, developing a deep understanding of the current trends in utilizing GAN-based adversarial attacks.
4) Sheds light on the future directions and challenges facing GAN-based adversarial attacks, which will help researchers contribute toward building a robust IDS.
5) Utilize BiGAN, Stackelberg GAN (SGAN), and CycleGAN to generate adversarial attacks against IDSs. BiGAN is used to influence the generation of malicious traffic by incorporating a latent representation learned from the original normal traffic. This approach enables the generation of more realistic and stealthy adversarial samples, making them harder to detect by IDSs. On the other hand, SGAN is used to address the training instability often observed in traditional GANs and to enhance the diversity of the generated adversarial samples. CycleGAN is particularly valuable for generating adversarial attacks because its architecture learn bidirectional mappings between domains, allowing a sample to be transformed from domain A to domain B and back again. This bidirectional capability, enforced by the cycle-consistency constraint, helps preserve the semantic content of the original sample while producing realistic-looking domain-shifted examples.
6) Examine 32 representative use cases. By applying GAN-based attacks against two up-to-date datasets, we show empirical evidence of GAN-based adversarial attacks' severity on IDSs under different scenarios.

Article Organization—The rest of the article is organized as follows. In Section II, we provide a comprehensive overview of the related reviews. Our methodology for systematic review is presented in Section III. In Sections IV and V, we review existing GAN-based adversarial attacks and defense strategies, respectively. A brief background on GAN-based attacks and defense strategies is also delivered in Sections IV and V, respectively. We provide meta-analysis on building the threat models and utilizing GAN-based adversarial attacks in Sections VI and VII, respectively. In Section VIII, we conduct a meta-analysis on severity assessments and the defense mechanisms used for GAN-based adversarial attacks. In Section IX, we deliver an in-depth discussion and future directions for GAN-based adversarial attacks. We examine 14 representative use cases to identify the effects of GAN-based adversarial attacks under different scenarios in Section X. Finally, in Section XI, we outline the main conclusion of this article.

II. Related Surveys

GANs are used in various applications, including advanced tasks that involve generating, enhancing, or manipulating images. Recently, GANs have been used in other domains, including cybersecurity, to generate adversarial attacks that deceive ML models. GAN-based adversarial attacks were extensively explored in computer vision (CV) and natural language processing (NLP), imposing significant threats against such systems. As a result, it became crucial to investigate such attacks in critical applications related to computer network traffic, which are presented in tabular data, such as IDSs. Protecting IDSs against adversarial attacks is crucial for defending critical systems that use IDSs. This study

TABLE I
COMPARISON OF RECENT SURVEYS

Study	Year	IDS	GAN	Adversarial attack	Adversarial Attack Metrics	Adversarial Defense	Adversarial Attack Challenges	Systematic	Use-Case
[17]	2023	High	None	None	None	None	None	✓	✗
[18]	2024	High	None	None	None	None	None	✓	✗
[13]	2023	None	Low	High	High	High	Non	✗	✗
[14]	2024	None	Low	High	High	High	Non	✗	✗
[12]	2023	High	Low	High	High	High	High	✗	✗
[15]	2023	High	Low	High	None	Low	Medium	✓	✗
[19]	2024	Medium	Low	High	High	High	High	✗	✗
[20]	2021	High	High	None	None	None	None	✗	✗
[10]	2021	Medium	High	Low	None	None	None	✗	✗
[21]	2023	None	High	None	None	None	None	✗	✗
[11]	2023	High	High	Low	None	None	None	✗	✗
[22]	2023	None	High	High	High	None	None	✗	✗
Our paper	2024	High	High	High	High	High	High	✓	✓

bridges the gap between GANs, adversarial attacks, and IDSs, providing a comprehensive analysis of GAN-based attacks against IDSs. Related surveys in the literature are focused either on IDSs, adversarial attacks, or GANs, as detailed in Table I.

A. IDSs

Utilizing artificial intelligence using various ML and DL techniques has made significant advancements in IDSs. Motivated by the lack of a dedicated study that systematically analyses the existing methodologies used in IDSs, Abdulganiyu et al. [17] provided a systematic review of the existing contributions of anomaly-, signature-, and hybrid-based approaches for IDSs in terms of the used datasets, types of attacks detected, evaluation metrics, and limitations of each contribution. The authors concluded that most of the existing works focused on ML-based IDSs despite the fact that DL approaches give the best performance. In addition, outdated datasets were the most used datasets, which raises serious concerns about the need for more updates and threats that were not considered in these datasets.

Satilmiş et al. [18] presented a systematic review focusing only on host-based intrusion detection systems (HIDS), which are IDSs for individual devices to monitor and analyze systems. The authors discussed the foundational approaches in HIDS in the context of the used datasets, intrusion detection method (signature-based or anomaly-based), evaluation criteria, and the advantages/disadvantages of ML/DL methods. The authors also pointed out several challenges that HIDS faces. The review emphasized the need for HIDS research to focus on up-to-date datasets, as well as real-world testing.

However, neither of these studies pointed out the risks of adversarial attacks against IDSs, which affect the reliability of such systems. Therefore, investigating IDSs in the context of adversarial attacks became crucial.

B. Adversarial Attacks Against IDSs

Adversarial attacks have been used in CV, NLP, and cybersecurity [12]. The concept of adversarial examples emerged as a significant concern as early as 2004 when Dalvi et al. [23] were able to modify spam emails to evade ML classifiers. This work was the groundwork for understanding how adversaries could exploit the weaknesses of ML models by making small changes to input data.

Because of the simplicity of the fast gradient sign method (FGSM) adversarial attacks proposed by Goodfellow et al. [24], the field of adversarial attacks gained significant momentum. Costa et al. [13] and Baniecki and Biecek [14] provided an extensive overview of the latest advancements in adversarial attacks, the metrics used, and the defense strategies against such attacks. However, they failed to cover systems such as IDSs. Motivated by this gap, the authors of [12], [15] explored the effects of adversarial attacks against IDSs, focusing on the gradient-based adversarial attacks against DL models. However, none of these surveys provides an extensive analysis of the effects of GAN-based adversarial attacks against IDSs.

C. GANs

GANs have been used since 2014 in various applications, including anomaly detection. A few reviews have been conducted on GAN applications. Sabuhi et al. [20] provided a systematic review of GAN applications in the field of anomaly detection in terms of the role of GANs, the commonly used architectures, and the metrics used to evaluate the performance of GANs. The survey papers [10], [11] addressed the use of GANs in cybersecurity. Navidan et al. discussed in [10] the use of GANs in network-related applications, including network slicing and self-organizing networks. On the other hand, Dunmore et al. [11] provided a detailed overview of using GANs as IDSs. They discussed the studied GAN types, datasets, and evaluation metrics. Nevertheless, [10], [11] overlooked GAN-based adversarial attacks, which can impose serious concerns on critical systems.

Zhang et al. [22] discussed the security and privacy of ML models against GAN-based adversarial attacks. Nevertheless, the authors did not cover the effects of such attacks on IDSs. The nature of GAN-based adversarial attacks can vary depending on the targeted system. Employing adversarial attacks on network traffic imposes greater challenges compared with other systems due to the semantic nature of network traffic. Modifying any portion of data can affect the validity of the remaining network traffic data.

To the best of our knowledge, no dedicated study has provided a systematic review focusing on GAN-based adversarial

TABLE II
SEARCH STRINGS

S1	((Generative adversarial network) OR (GAN)) AND (adversarial attack) AND ((intrusion detection system) OR (IDS) OR (NIDS))
S2	((Generative adversarial network) OR (GAN)) AND ((adversarial attack) OR (adversarial machine learning)) AND ((intrusion detection system) OR (intrusion detection systems) OR (IDS) OR (NIDS))
S3	((Generative adversarial network) OR (GAN)) AND ((adversarial attack) OR (adversarial machine learning) OR (adversarial example)) AND ((intrusion detection system) OR (intrusion detection systems) OR (IDS) OR (NIDS))

attacks against IDSs. Therefore, this article aims to bridge this gap by delivering an in-depth review of GAN-based attacks in terms of threat model development, adversarial attack generation, the assessment of threat models under such attacks, and the strategies employed to defend IDSs against them. Beyond offering a general survey, we have conducted a meta-analysis of the key influencing factors that affect the process of developing threat models, generating GAN-based attacks, and assessing and defending the threat models. Furthermore, we consolidated our findings through 14 use-case studies.

The key differences between our review and earlier surveys are.

1) Our review focuses solely on GAN-based attacks, whereas prior surveys tend to cover adversarial attacks more broadly without differentiating based on the attack generation method.
2) We introduce specialized severity metrics tailored to evaluating GAN-generated adversarial samples, such as the Fréchet inception distance (FID) and the Hellinger distance (HD), which go beyond traditional evaluation metrics such as accuracy or evasion rate and better capture the stealthiness and realism of adversarial examples.
3) Through empirical analysis, we examine how feature selection strategies influence attack success rates and model robustness, providing a nuanced understanding that was largely absent from earlier reviews.

III. METHODOLOGY

In this section, we present our implementation of the PRISMA protocol to identify the set of papers that will be comprehensively reviewed in the following three sections. The review questions that will be used in reviewing these papers are also explained in this section.

A. Search Queries and Strings

This section presents the search queries and strings used to collect papers related to GAN-based adversarial attacks against IDSs. The keywords used for the initial search are GAN, adversarial attacks, and IDS. Using these keywords, Table II shows three search queries used to collect the papers.

B. Inclusion and Exclusion Criteria

1) Inclusion Criteria: Every paper that satisfies the following two criteria is included.

IC1: Conference proceedings and journal articles.

IC2: Papers that use GAN-based adversarial attacks on IDSs.

2) Exclusion Criteria: Every paper that satisfies one of the following criteria is excluded.

EC1: A short paper, a letter, or a comment paper. These formats typically lack comprehensive methodological details and experimental evaluations, making them unsuitable for an in-depth systematic analysis.

EC2: Papers used datasets collected or released before 2010.

EC3: A survey or a systematic review. Our objective was to analyze primary research contributions rather than resummarizing existing reviews.

EC4: Papers that used GANs for data augmentation were excluded to maintain the focus of the study on adversarial threat modeling.

EC5: Papers that used GANs as IDSs rather than adversarial attack generators were excluded, as their primary objective does not align with evaluating vulnerabilities in IDSs.

EC6: A study that lacks sufficient methodological detail to the extent that it is not reproducible.

C. Quality Assessments

Papers resulting from the inclusion and exclusion criteria went through the quality assessment process, where they were carefully assessed based on the following criteria.

QA1: Does the study evaluate IDSs using different classifiers and/or datasets? If the paper considers multiple IDSs/datasets, then the paper score is 1; otherwise, it receives a score of 0.

QA2: Does the study report results with comparison to baseline methods? If the study includes a comparison with GAN-based adversarial attack baselines, it is awarded a score of 1. If the comparison is made with gradient-based or other types of adversarial attacks, it is given a score of 0.5. In cases where no comparison to any baseline methods is provided, the study receives a score of 0.

QA3: Does the paper propose a new GAN-based adversarial attack or enhance an existing one? If the study introduces a novel GAN-based adversarial attack or presents a meaningful enhancement to an existing approach, it receives a score of 1. Enhancements may include the use of different GAN architectures or modifications in the way features are manipulated. If no such contribution is made, the study receives a score of 0.

The total score of a paper is the sum of the scores obtained in QA1, QA2, and QA3. Research papers scored below 2 are excluded. Papers included (scored 2 or above) provide at least one substantial methodological component, such as multidataset/IDS evaluation, existing GAN-based attack enhancement/new GAN-based attack development, or baseline comparison. Papers providing comparative studies are excluded if scored below 1.5. Comparative studies included (scored 1.5 or above) utilize multiple datasets/IDSs and provide a baseline comparison.

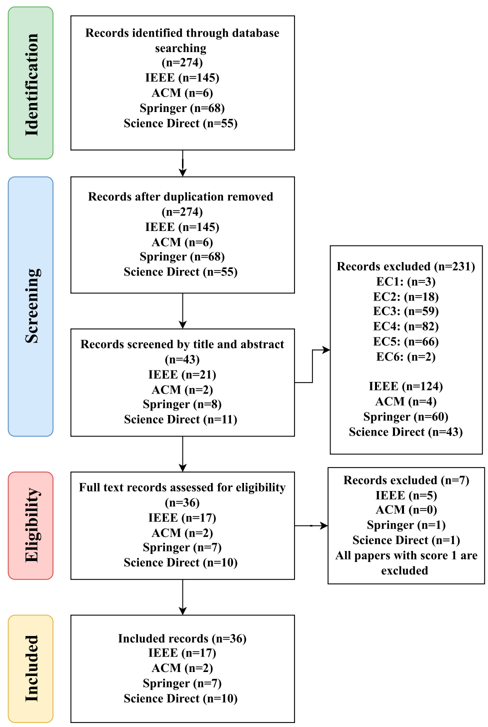

Fig. 2. PRISMA diagram.

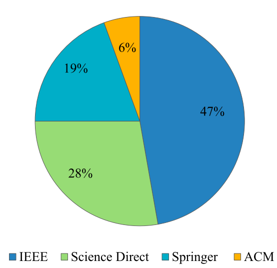

Fig. 3. Paper distribution based on the sources.

D. PRISMA Diagram

Fig. 2 illustrates the PRISMA diagram that summarizes our systematic review process. The search queries shown in Table II were used to retrieve all 274 papers from four different databases: 1) *IEEE*; 2) *ACM*; 3) *Springer*; and 4) *Science Direct*. Then, it was confirmed that there were no duplicate papers. After applying the inclusion and exclusion criteria, defined in Section III-B, on these papers, three papers were excluded due to EC1, 18 due to EC2, 59 due to EC3, 82 due to EC4, 66 due to EC5, and two due to EC6. The remaining 42 papers went through the quality assessment process, and six papers were excluded.

Fig. 3 shows that *IEEE* is the most contributing database to the field of GAN-based adversarial attacks against IDSs, followed by *Science Direct*. Fig. 4 represents the growing rate of publications in each database across the last 4 years.

Next, we determine a set of questions for reviewing the final list of 36 papers. These papers will be comprehensively

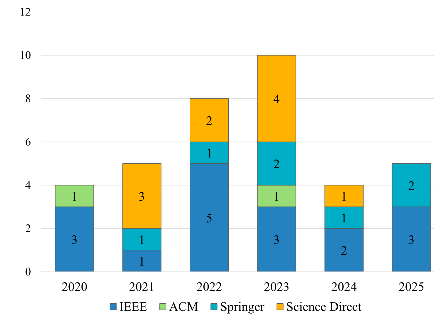

Fig. 4. Annual publication trends from 2020 to 2025 across different databases.

reviewed based on these questions in Sections IV and V. Our set of review questions can be divided into three subsets. 1) threat model review questions; 2) adversarial attack generation review questions; and 3) security assessment and defense mechanism review questions.

E. Threat Model Review Questions

This section presents the review questions regarding the process of developing the threat (target) model.

RQ1: What are the DL or ML techniques used in IDSs (threat models)?

A1: This question can give insights into the robustness of the models against adversarial attacks. Models can be built using ML or DL algorithms, such as artificial neural networks (ANNs), convolution neural networks (CNNs), auto-encoders, recurrent neural networks (RNNs), long short-term memory (LSTM) networks, gated recurrent unit (GRU) networks, residual neural networks (ResNets), decision trees (DTs), support vector machines (SVMs), K-nearest neighbors (KNNs), naive based (NB), random forest (RF), linear regression (LR), adaptive boosting (AdaBoost), extreme gradient boosting (XGBoost), transfer learning (TF), autoencoders (AEs), and others.

RQ2: What is the type of the IDS classifier?

A2: The type of classifier can affect the overall performance of the IDS, which in turn affects the robustness of the classifier against adversarial attacks. There are two main types of classifiers: binary and multiclass.

RQ3: What are the datasets used to train the IDS?

A3: The used datasets can directly affect the IDS performance. Many datasets can be used to train the IDS model, such as CICIDS2017, CICIDS2018, CICDDoS 2019, UNSW-NB15, Kitsune, IoT-23, Bot-IoT, CCD-INID-V1, CCDIDSV1, IEC, virusTotal (VT), DAPRA, ADFA-LD, Hyundai YF Sonata, UNB-CIC, Drebin, ISOT, X-IIoTID, MAWILAB, and private datasets (their own).

F. Adversarial Attacks Generation Review Questions

In this section, we identify all review questions related to the process of generating GAN-based adversarial attacks.

RQ4: What is the type of GAN used?

A4: The type of GAN can be determined by one of the following: Wasserstein GAN (WGAN), GAN (Vanilla), conditional GAN (cGAN), conditional tabular GAN (CTGAN), least-square GAN (LSGAN), PT-GAN, information-theoretic GAN (InfoGAN), few feature attack GAN (FFA-GAN), deep convolutional GAN (DCGAN), and bidirectional GAN (BiGAN).

RQ5: What is the game model used in GAN?

A5: According to [21], different types of game models can be used in GANs, such as zero-sum games, stochastic games, and Stakelberg games.

RQ6: What is the underlying architecture used in GAN?

A6: GANs can have the following architectures: One-generator and one-discriminator, multigenerator and one-discriminator, one-generator and multidiscriminator, multigenerator, and multidiscriminator.

G. Severity Assessment and Defense Mechanism Questions

The review questions regarding the process of assessing and defending the threat model are presented in this section.

RQ7: What are the metrics used to analyze the effectiveness of GANs?

A7: Many metrics can be used to assess the quality of the data generated from GANs. This includes the inception score (IS), Fréchet inception distance (FID), GAN loss, correlation, Jaccard, Kolmogorov-Smirnov test (KS-test), and Hellinger distance.

RQ8: What are the metrics used to analyze the effectiveness of adversarial attacks against IDSs?

A8: As the main goal of the adversarial attacks is to mislead the IDS, there are many ways to measure the performance of IDSs under GAN-based adversarial attacks, namely accuracy, recall, precision, F1-score, false positive rate (FPR), true positive rate (TPR), detection rate, and evasion rate.

RQ9: How do existing IDSs defend against GAN-based adversarial attacks?

A9: Defending IDSs against adversarial attacks can be done in several ways: Adversarial training, auto-encoder, or one-class SVM.

IV. Review of Adversarial Attacks

In this section, we present a brief background on GAN-based adversarial attacks on IDSs and review related literature.

As mentioned earlier, adversarial attacks can be categorized based on the attacker's knowledge, goal, and algorithm, as presented in Fig. 1. Based on the attacker's knowledge, adversarial attacks can be white-box, black-box, or grey-box attacks. White-box attacks assume that the attacker has full knowledge of the system, the dataset used, and all aspects of the IDS model. In black-box attacks, the attacker is assumed to have almost no knowledge of the IDS model. Grey-box attacks assume a wide range of possibilities. Some grey-box attacks assume that the attacker does not have all the information about the IDS. Other grey-box attacks assume that the type of the dataset used may not be known. There also exist grey-box attacks that assume the type of IDS may be known without being aware of all its hyperparameters, as explained in [12].

Regarding the attacker's goal, adversarial attacks can be targeted or untargeted. In the untargeted attack, the attacker's main goal is to make the classifier misclassify the adversary sample. On the other hand, attackers in the targeted attack aim to deceive the classifier into classifying an adversary sample into a specific class, as noted in [12].

Adversarial attacks can be generated using different algorithms. Gradient-based methods rely on calculating the gradient of the loss function with respect to the input data, as mentioned in [24]. By adjusting the input in the direction of the gradient, these methods generate adversarial examples that can lead to misclassification. However, this method can only target models that use gradients, such as the DL methods, as stated in [25]. Moreover, these types of attacks are considered white-box attacks since the attacker must have access to the model to know the gradient of a specific sample. The FGSM proposed in [24] is the most common gradient-based adversarial attack.

Metaheuristic approaches such as genetic algorithms and particle swarm optimization can also be used to generate adversarial attacks by using optimization models inspired by natural processes, as experimentally proved in [26], [27]. These methods can be used on black-box attacks, where the attacker explores the solution space globally to find the optimal perturbation that causes misclassification.

In reinforcement-based attacks, the attacker (agent) is trained using reinforcement learning to generate adversarial examples. The agent receives feedback from the model (rewards or penalties) based on its success in causing the model to make incorrect predictions, and it adjusts its strategy accordingly. This type of attack is a black-box attack where the attacker has no knowledge about the system model [28].

GAN-based adversarial attacks assume that the attacker has no prior knowledge of the threat model in terms of the type and nature of the classifier. The attacker also uses GANs to learn the distribution of normal data and then produce new adversarial samples that appear similar to the normal data. Moreover, GAN-based attacks can also be seen as targeted attacks that aim to classify abnormal traffic as normal traffic.

Generating GAN-based adversarial attacks requires a threat or a target model to be attacked, and mainly, there are five steps to be considered when developing a threat model, as shown in Fig. 5. Building a threat model starts with the dataset selection, a crucial step that directly affects the model's performance. Datasets with good data quality and large diversity can be used to build an effective model with high performance.

After selecting the dataset, the preprocessing step will take place, where the dataset will be prepared to be used as input for the model. This step involves converting all categorical features into numeric features, cleaning the data, removing duplication, removing unnecessary features (feature selection), and normalizing the values as needed. This step is considered pivotal in determining the model type, whether a binary classification or a multiclassification model.

After completing the steps related to the dataset, the process of defining and building the model begins. This is done by determining whether the model will be built using ML or DL. The choice between these two options depends on the size and nature of the data, as well as the available computing power. If the data size is large, its nature contains complex patterns, and there is sufficient computing power, then DL will be a better choice. The selection process often involves trial and error, aiming to identify the model that delivers the best performance. Alternatively, a combination of models may be employed to enhance overall system effectiveness.

Once the model is determined, training the model will start by feeding the model with training data to capture and learn the patterns from the input features and corresponding labels. Once the model is trained, the testing phase will evaluate the model's performance on a separate part of the dataset that the model has never seen before. This step is critical to assess the model's generalizability to new, unseen data. During the testing phase, GAN-based adversarial scenarios should be simulated. This is called an evasion attack, as defined in [12], which aims to test the model's resilience against different types of GAN-based adversarial attacks.

GAN-based attacks are attacks crafted using a GAN, where the GAN architecture consists of two network generators responsible for creating the attacks and a discriminator that simulates the threat model. The input data to a GAN is divided into normal and malicious. The malicious records are passed to the generator. The generator learns to produce a noise vector; when added, the malicious records will fit under the normal traffic distribution, as proven in [29]. The noise vector size is determined based on the number of nonfunctional features of the traffic, which are characteristics of traffic that do not change the core functionality or harmful intent of an attack, nor the validity of the traffic, but can still be manipulated by an adversary to avoid detection systems. The intrinsic and functional features of the attack should both remain unchanged to preserve the validity of the traffic. The generated nonfunctional features are seamlessly integrated with the original data using a masking mechanism. Specifically, a binary mask is employed to distinguish between functional and nonfunctional (generated) features. This mask guides the fusion process by allowing only the nonfunctional features to be updated. This ensures that the functional features remain untouched, preserving the semantics of the original data.

The discriminator is trained to mimic the targeted model by taking the actual labels of the adversarial malicious records and the normal ones from the targeted model and using them for its training process. During the training process of a GAN, the generator tries to minimize its loss to generate adversarial samples that resemble the normal sample. In other words, the generator minimizes the expected value of the discriminator's output when evaluating the adversarial examples generated by the generator [29]. On the other hand, the discriminator continuously refines its capacity to differentiate between real and generated data. This adversarial process continues until an equilibrium is reached.

There are various types of GAN-based attacks, each characterized by its unique loss function and architecture. Vanilla GAN, for example, uses binary cross-entropy as a loss function, whereas WGAN, proposed in [30], uses the Wasserstein loss. On the other hand, LSGAN, presented by [31], adopts the least square loss function. The GAN architecture can also vary depending on the network used by the generator and discriminator. For example, DCGAN, proposed by [32], employs a deep CNN. The input structure to the generator can also influence the type of GAN. If the generator receives both a condition and a noise vector as input, the model is referred to as a cGAN [33]. When applied to tabular data, this variant is known as CTGAN [34]. Other types of GANs, such as BiGAN proposed in [35], go a step further by not only mapping a noise vector to generated data but also learning the inverse mapping, enabling the generator to infer input from the output data effectively.

Since there are many different ways of generating GAN-based adversarial attacks, as shown in Fig. 6, we categorize the previous studies into three classes: 1) GAN-based attacks; 2) WGAN-based attacks; and 3) other types of GAN-based attacks. Each category was further classified based on the type of the target model (binary or multiclass). Tables III and IV review the papers that use GAN-/WGAN-based attacks against binary threat models, respectively. From both tables, we can say that most of the existing studies targeting binary IDSs employ relatively simple GAN architectures, often introducing only minor modifications to the generator network, as demonstrated in [36], [37]. This approach highlights the inherent vulnerabilities present in binary IDSs. WGAN is preferred because it enhances stability, avoids mode collapse, and produces more realistic, stealthy adversarial samples, all critical for attacking IDSs effectively. Consequently, some studies propose additional constraints on the validity and use adversarial training as a defense mechanism against such attacks. Papers with other types of GAN-based attacks targeted binary models are summarized in Table V. As for multiclass models, both WGAN- and GAN-based attacks are presented in Table VI. Table VII summarizes the papers that use different architectures for GAN-based attacks targeting multiclass IDSs. When targeting multiclass IDSs, we observed that the literature introduces additional modifications due to the increased difficulty in bypassing all decision boundaries. For instance, some approaches incorporate spatial representation in adversarial sample generation [38], while others utilize projection-based techniques [39]. Due to the high possibility of encountering one of GAN's common problems (stability and mode collapse), especially when there are multiple attacks to be modified as normal ones, the authors of [40] used a fuzzy test technique to enhance the convergence.

V. Review of Defense Mechanisms

Once the adversarial attacks have been generated, it is crucial to evaluate their impact on the threat model. Understanding how the threat model is affected under attacks makes it easier to understand the system's vulnerabilities and propose a defense mechanism accordingly.

TABLE III
GAN-Based Adversarial Attacks Against Binary Threat Models

Study	RQ1	RQ3	RQ9	Main Findings
[36]	DNN	CICIDS2017	None	• Generated adversarial samples using a new algorithm, Gen-adversarial attacks, which is built upon GAN and active learning. • GAN generator network was built using a variational auto-encoder. • Achieved 98.86% attack success rate.
[41]	ANN, RF, LR	Hyundai YF Sonata	None	• Developed an in-vehicle IDS. • Generated adversarial samples against the developed IDS. • Achieved more than 50% degradation of performance.
[42]	DNN	UNB-CIC	None	• Proposed a new IDS that identifies Tor from non-Tor traffic. • Applied GAN-based adversarial attacks against the IDS. • The model accuracy was degraded from 98.97% to 73.02%.
[43]	SVM, RF, KNN, ANN, CNN, LSTM	Their own	None	• Built a bus-traffic IDS. • Generated GAN-based adversarial attack samples. • Applied adversarial training to increase the robustness of IDSs.
[37]	–	CICIDS2017	None	• Proposed CVAE-AN, a conditional variational auto-encoder adversarial network, which is a semi-supervised GAN.
[5]	ANN, DT, LR, RF, SVM, KNN, NB, Adaboost, GB, CNN	CCD-INID-V1, CCDIDSV1	None	• Proposed AIGAN, a GAN-based adversarial attack against IoT datasets.
[45]	LR, KNN, DT, RF, AdaBoost, CNN+LSTM	UNSW-NB15, CICIDS2017	None	• Developed AIDAE, an anti-intrusion detection auto-encoder. • Used GAN in AIDAE for learning the prior distribution of the latent space of the dataset.
[46]	CNN, RF	Their own	Auto-encoder	• Proposed a new data representation. • Generated adversarial attacks using GAN.
[47]	–	CICIDS2017	Adversarial training	• Presented a new way to generate a large number of adversarial attack samples from a single attack sample using GAN.
[48]	DT, RF, SVM, ELM, XGBoost, CNN, DNN, Stacking Classifier, AE	Drebin	None	• Utilized Android anti-malware models. • Explored the vulnerabilities of these models under GAN-based adversarial attacks.
[49]	CNN, LSTM, ANN	CICDDoS2019	Adversarial training	• Designed GAN-based adversarial attacks targeting SDN. • Used these attacks for adversarial training purposes.

TABLE IV
WGAN-Based Adversarial Attacks Against Binary Threat Models

Study	RQ1	RQ3	RQ9	Main Findings
[50]	RF, DT, LR, ANN, NB	ADFA-LD, DREBIN	None	• Developed BFAM, a brute-force attack method, which is a gradient-free method to generate adversarial samples. • Compared the severity of BFAM and WGAN adversarial attacks.
[51]	NB, LR, RF, DT	CICIDS2017	None	• Used WGAN to launch distributed denial-of-service (DDoS) attacks by updating the feature profiles to generate undetectable samples.
[52]	CNN	CICIDS2017, UNB201X	Adversarial training	• Proposed GADoT, a WGAN-based adversarial attack with gradient penalty targeting DDoS-based IDSs.
[53]	SVM, NB, RF, ANN, KNN, DT, SVM, CNN, RNN	CICIDS2018	None	• Presented DIGFuPAS, a WGAN-based adversarial attack that preserves the functional features of the adversary samples.
[54]	ANN, LR, KitNet	Kitsune, CICIDS2017, MAWILAB, UNSW-NB15	Adversarial training	• Proposed FPAttack, a packet-level adversarial attack using WGAN. • Presented four packet insertion methods.
[55]	LR, DT, CNN, LSTM, ANN	CICIDS2018	None	• Studied the inconsistencies in adversarial samples and the validity of these samples in the context of value ranges, binary values, and membership in multiple categories.
[56]	DT, ANN, KNN	InSDN, CICIDS2017	Auto-encoder	• Presented RAIDS, a robust IDS against feature-level WGAN and CTGAN-based adversarial attacks.
[57]	SVM, DT, RF, KNN, CNN, GRU, CNN + GRU	X-IIoTID	Adversarial training	• Proposed EIFD, a framework for building a robust IDS against different types of attacks, including WGAN-based attacks.
[58]	DT, RF, ANN, LSTM, XGBoost	CICDDoS2019, CICIDS2017	Adversarial training	• Proposed a new IDS using LSTM. • Generated WGAN-based attacks against LSTM-based IDSs.
[59]	KNN, RF, SVM, LR	CICIDS2017, CICIDS2018	Adversarial training	• Proposed DA-GAN for generating mutated network traffic and preserving their functionality. • Generated domain-adaptive adversarial attacks by identifying a shared latent subspace between domains.
[60]	IF, SAE, SVM, GMM, RBM	DARPA, Kitsune, CICIDS2017	None	• Presented MACGAN, a framework for attacking IDSs and preserving the traffic validity.
[61]	DT, RF, ANN	ISOT	Adversarial training	• Established a fine-tuned IDS using metaheuristic algorithms. • Generated GAN-based adversarial attacks and tested their transferability against different models.
[62]	ANN	CICIDS2017	None	• Introduced a robust IDS using hierarchical packet attention convolution systems and tested the robustness of the proposed system under GAN- and gradient-based adversarial attacks.
[63]	RNN, LR, MLP	CICIDS2017	Honeypot IDS	• Presented S-IDS, a honeypot IDS that strategically flips the labels of network flows, marking benign traffic as malicious and vice versa to disrupt the attacker's model training process.

TABLE V
OTHER TYPES OF GAN-BASED ADVERSARIAL ATTACKS AGAINST BINARY THREAT MODELS

Study	RQ1	RQ3	RQ4	RQ9	Main Findings
[64]	DT, LR, AB, GB, KNN, SVM	Their own	LSGAN	Adversarial training	• Proposed LSGAN-AT, a framework that produces more effective adversarial malware samples using LSGAN.
[65]	DT, MLP, LR, XGBoost	CICIDS2017	FFA-GAN	None	• Used a new GAN architecture for generating adversarial attacks by modifying a few network traffic features.
[66]	MLP, KNN, SVM, LSTM, CNN, RF	Their own	CWGAN	None	• Utilized CWGAN-based adversarial attacks that can deceive IDS in cyber-physical systems (CPS) with minimal alterations to the input data.
[67]	ST, RF, LR, SVM	AndroZoo, CICAnd-mal2017	Auto-encoder and transformer based GAN	None	• Investigated the efficacy of transformer-based versus auto-encoder-based GANs for generating adversarial samples against IDS and mobile malware detection.
[68]	CNN, TF, AE, GAAD	Their own	U-shape GAN	None	• Proposed a combination of GANs with U-shaped network architectures, such as U-Net, to generate adversarial examples against UAV.
[69]	SVM, NB, DT, RF, LR	CICIDS2017	LSTM GAN	None	• Proposed a novel framework that combines a self-attention mechanism with a GAN to generate adversarial network packets.

TABLE VI
WGAN-BASED AND GAN-BASED ADVERSARIAL ATTACKS AGAINST MULTI-CLASS THREAT MODEL

Study	RQ1	RQ3	RQ4	RQ9	Main Findings
[38]	KNN, RF, CNN, XGBoost, LSTM, TF	CSE-CIC-IDS2018, CICIDS2017, UNSW-NB15	WGAN	None	• Proposed ProGen, a new framework for generating adversarial attacks using a projection-based technique of the basic feature sequence.
[39]	3L-IDS, LUNET, CNN+LSTM, SVM, LR, RF, KNN, DT	UNSW-NB15, CICDDoS2019	WGAN	None	• Proposed NIDSFM, a framework that uses the latent spatial representation for benign samples. • NIDSFM ensures that the adversarial sample has the same benign latent representation.
[40]	ANN, RNN, LSTM, DNN, GRU	CICIDS2017	WGAN	None	• Proposed VulurGAN, a timeliness backdoor attack on IDSs to poison the data and to launch evasion attacks. • Used a fuzzy test to enhance the convergence of VulurGAN.
[70]	DT, RF, SVM	CSE-CIC-IDS2018	GAN	Adversarial training	• Used GAN-based attacks for training, to increase the robustness of IDSs against the Carlini-Wagner attack.
[71]	LR, NB, SVM, LSTM, KNN	CICIDS2017	EGAN	Adversarial training	• Harnessed the self-attention mechanisms and GANs to enhance the resilience of IDSs.

TABLE VII
OTHER TYPES OF GAN-BASED ADVERSARIAL ATTACKS AGAINST MULTI-CLASS THREAT MODELS

Study	RQ1	RQ3	RQ4	RQ9	Main Findings
[72]	DT, RF, XGBoost, ANN	IEC	CTGAN	None	• Developed CTGAN-based adversarial attacks and FGSM attacks.
[72]	CNN, DNN	UNSW-NB15	PT-GAN	OneClass-SVM	• Used PT-GAN for poisoning the training dataset.

The defense mechanism aims to mitigate the impact of adversarial attacks and increase the robustness of the threat model. One common approach is adversarial training, where the threat model tries to distinguish between normal data and adversarial samples in the training phase, as experimentally proven in [24]. A similar approach to adversarial training is using a secondary binary classifier that classifies the adversary sample from the original ones. These two approaches enable the models to recognize and resist such attacks. Adversarial training, however, is computationally expensive and has low scalability, making it more suitable for static and resource-rich environments. Another approach is input data preprocessing or sanitization, where the input is transformed or cleaned before being fed into the model. This preprocessing step can include techniques such as noise reduction [45], [56], [73] or feature squeezing [74]. The goal of these methods is to mitigate the adversarial perturbations that have been introduced into the input data. By doing so, the model receives a "cleaner" version of the input, reducing the likelihood that adversarial manipulation successfully misleads the model. Preprocessing techniques are lightweight, highly scalable, and easy to deploy, making them well-suited for real-time systems where low computational overhead is crucial. This type of defense is particularly effective because it does not require changes in the model architecture, making it relatively easy to integrate into existing systems.

Although model ensembling is considered to be a more scalable approach than adversarial training and more robust than data preprocessing, relying solely on an ensemble of models is not sufficient in the context of IDSs, particularly against GAN-based adversarial attacks, due to the high transferability of such attacks. In adversarial ML, transferability refers to the ability of adversarial examples, crafted to fool one model, to successfully deceive other models, even if they differ in architecture or training data. In traditional ensemble methods,

TABLE VIII
COMPARISON BETWEEN DEFENSE MECHANISMS

Aspect	Adversarial Training	Ensemble Classifier	Preprocessing
Scalability	Low	Moderate	High
Computational cost	High	Moderate	Low
Real-world applicability	Static and resource-rich environments	Cloud-based systems	Real-time systems
Robustness	Strong (to known attacks)	Moderate to strong	Weak to Moderate

the assumption is that diversity among classifiers can make it harder for an adversarial input to simultaneously fool all components. However, GANs can learn to generate perturbations that target shared vulnerabilities across different models, thereby preserving their transferability and undermining the robustness of ensemble-based defenses.

The Apollon framework [75] is an effective ensemble model that enhances the robustness, which directly incorporates this insight by employing a robust ensemble of heterogeneous models, each trained using different learning algorithms and feature sets. By using a multiarm bandit, the architectural and functional diversity among ensemble components, Apollon significantly reduces the likelihood that a single adversarial perturbation will be effective across all models. Using such a framework not only improves overall detection accuracy but also mitigates the risk posed by transferability. By leveraging a well-designed ensemble of classifiers, the IDS can be fortified against both white-box and black-box adversarial attacks. The authors of [62] put forth a new defense mechanism against GAN-based adversarial attacks. The main goal in their work is to subvert the training process of GANs by integrating a honeypot IDS that strategically flips the labels of network flows, marking benign traffic as malicious and vice versa, to disrupt the attacker's model training process. In Table VIII, we present a summary of existing defense mechanisms against GAN-based adversarial attacks.

In the context of defending IDSs, the most common approach against GAN-based adversarial attacks is adversarial training. In Fig. 7, papers are categorized based on the used defense strategy into adversarial training and other approaches (including data preprocessing and using secondary classifiers). Table IX presents the papers that used adversarial training as a defense mechanism against GAN-based attacks. Additionally, Table X summarizes the main findings of the papers that implement different types of defense mechanisms against GAN-based attacks. In the following sections, we provide a meta-analysis, a method combining data from multiple independent studies centered on a common research question. It offers a comprehensive overview and highlights common trends by addressing and examining the effects of the predefined review questions.

VI. THREAT MODEL DEVELOPMENT

The process of understanding and identifying any potential threats and vulnerabilities starts with building a comprehensive threat model. This section outlines the method used for building the threat model in the literature. To prepare for potential adversarial attacks, it is crucial to develop a comprehensive threat model. Fig. 5 illustrates the key steps for building any IDS. The following sections discuss the review questions regarding each step.

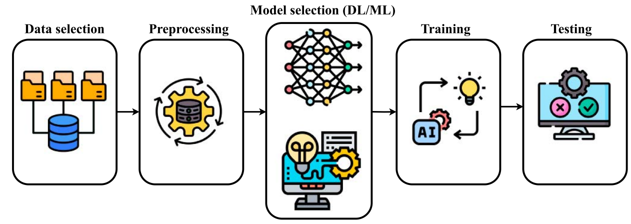

Fig. 5. Main steps for developing an IDS threat model.

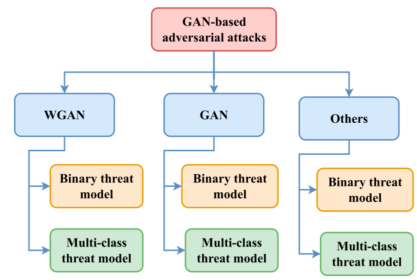

Fig. 6. Taxonomy of GAN-based attacks.

TABLE IX
ADVERSARIAL TRAINING AS A DEFENSE MECHANISM AGAINST W/GAN-BASED ADVERSARIAL ATTACKS

Study	RQ1	RQ2	RQ3	RQ4
[70]	DT, RF, SVM	Multi-class	CSE-CIC-IDS2018	GAN
[59]	KNN, RF, SVM, LR	Binary	CICIDS2017, CICIDS2018	WGAN
[49]	CNN, LSTM, ANN	Binary	CICDDoS 2019	WGAN
[54]	ANN, LR, KitNet	Binary	Kitsune, CICIDS2017, MAWILAB, UNSW-NB15	WGAN
[64]	DT, LR, AB, GB, KNN, SVM	Binary	Their own	LSGAN
[47]	–	Binary	CICIDS2017	GAN
[58]	DT, RF, ANN, XGBoost, LSTM	Binary	CICDDoS 2019, CICIDS2017	WGAN
[57]	SVM, DT, RF, KNN, CNN, GRU, CNN+GRU	Binary	X-IIoTID	WGAN
[61]	DT, RF, ANN	Binary	ISCX, ISOT	WGAN

A. Dataset Selection

In this step, the dataset is selected based on the scope in which the IDS will be built. Generally, there are five main categories: 1) IoT; 2) network traffic; 3) malware analysis; 4) automotive cybersecurity; and 5) others. Most researchers who studied the effects of adversarial attacks in the past four years built their threat model based on network traffic, as illustrated in Fig. 8.

TABLE X
OTHER TYPES OF DEFENSE MECHANISMS AGAINST W/GAN-BASED ADVERSARIAL ATTACKS

Study	RQ1	RQ2	RQ3	RQ4	RQ9	Main Findings
[73]	CNN, DNN	Multi-class	UNSW-NB15	PT-GAN	OneClass-SVM	• Presented a novel defense mechanism for poisoning data detection. • Developed a layer-wise relevance propagation for extracting important neurons.
[46]	CNN, RF	Binary	Their own	GAN	Auto-encoder	• Proposed a new data representation. • Generated adversarial attacks using GANs. • Used a denoising auto-encoder to reconstruct the adversarial samples.
[63]	RNN, LR, MLP	Binary	CICID-S2017	WGAN	Honeypot IDS	• Presented S-IDS, a honeypot IDS that strategically flips the labels of network flows, marking benign traffic as malicious and vice versa to disrupt the attacker's model training process.

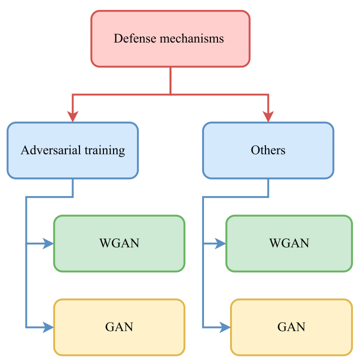

Fig. 7. Taxonomy of defense mechanisms.

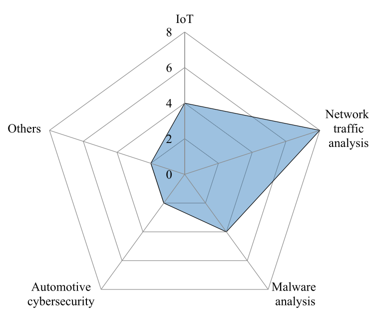

Fig. 8. Distribution of the papers on the used dataset categories.

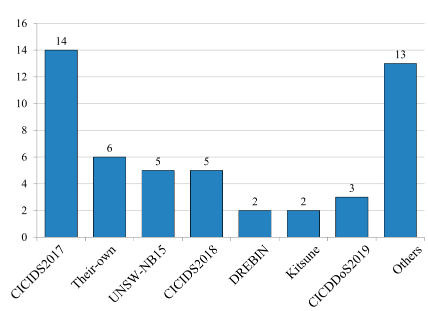

Fig. 9. Frequency of the used datasets in the literature.

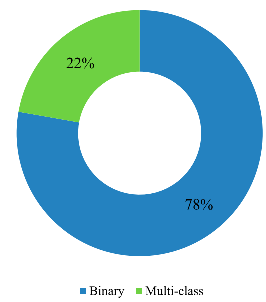

Fig. 10. Ratio of binary and multiclass models.

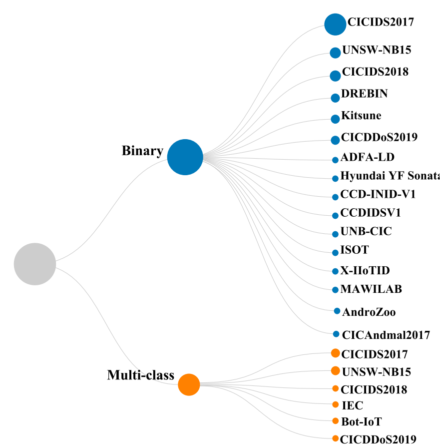

Fig. 11. Frequency of the types of classifiers used in the datasets.

The main reason for such a trend is that network traffic datasets are considered more general than others. On the other hand, the least used datasets were automotive cybersecurity, due to the lack of public datasets of such type.

The process of selecting the dataset represents the effect of *RQ3*. Fig. 9 demonstrates the frequencies of studies considered for each dataset. Throughout the last four years, there have been many datasets to be used in IDSs, and the most used dataset was CICIDS2017, followed by UNSW-NB15 and CICIDS2018 datasets. The other category represented in the last column includes the following datasets: ADFA-LD, AndroZoo, CICAndmal2017, CCD-INID-V1, CCDIDSV1, IEC, ADFA-LD, Hyundai YF Sonata, UNB-CIC, ISCX, ISOT, X-IIoTID, and MAWILAB. Each one of these datasets contributes to the "others" category by one. It can be noticed that the effect of GAN-based adversarial attacks was effectively studied in the CICIDS2017 dataset for two main reasons: 1) it has become a widely accepted benchmark within the research community, because of its early release; and 2) CICIDS2017 also enables better reproducibility and cross-study comparison, as many existing works have established performance baselines using this dataset. However, the nature of attacks may differ

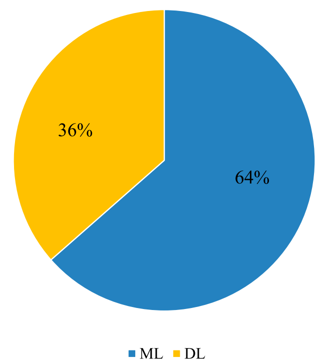

Fig. 12. Ratio of ML to DL models.

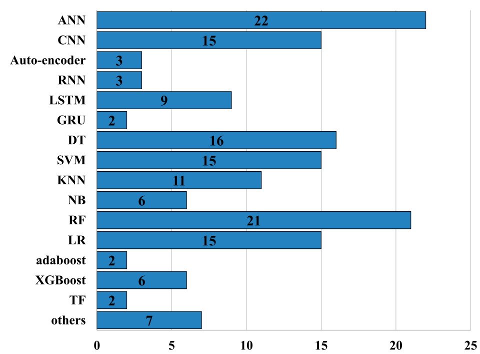

Fig. 13. Distribution of various ML and DL models.

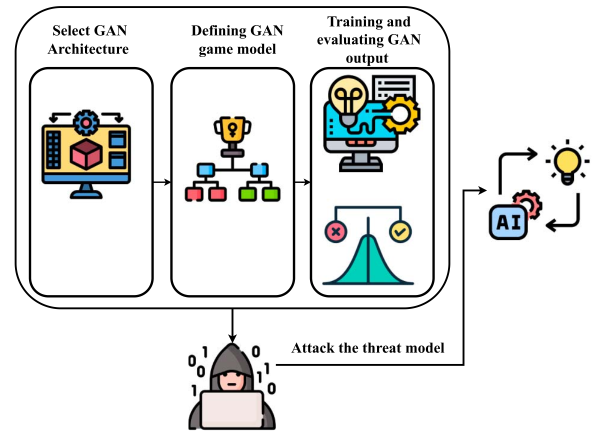

Fig. 14. Main steps for generating GAN-based adversarial attacks.

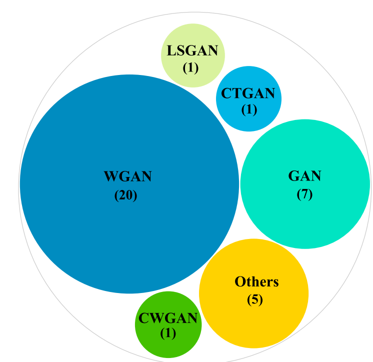

Fig. 15. Types of GAN used in the literature.

from one dataset to another. It is crucial to see whether the effectiveness of a GAN-based adversarial attack depends on the used dataset in the threat model or not. This will be further discussed in Section VII.

B. Preprocessing

After selecting the dataset that will be used for training the threat model, the preprocessing step begins. The most essential part of this step is to convert all categorical features into numeric features, including the class or the label. There are two ways to deal with the class, either by converting it to a binary or multiclass, which, in turn, affects the classifier type. The type of classifier can significantly affect the complexity of the adversarial attack. In binary classification, there is only one decision boundary, and the goal of the adversarial attack is to push a data point across that decision boundary. The multiclassification scenario requires more complex decision boundaries. Crafting adversarial examples in this context is more challenging because the attacker must navigate more than one decision boundary and consider multiple classes. As a result, 78% of researchers studied the impact of adversarial attacks using binary classifiers, as shown in Fig. 10.

The nature of the dataset and its features may limit the direction to a particular classification type, which determines the way attacks are created. Fig. 11 represents the distribution of the classifier types used in the datasets across the literature.

It can be noticed that the impact of adversarial attacks has been studied in some datasets in both scenarios (binary and multiclassification), such as CICIDS2017, UNSW-NB15, CICIDS2018, and CICDDoS2019. On the other hand, the impact of GAN-based adversarial attacks has been studied under one type of classifier for certain datasets, and this can open up a new direction to be discovered, as will be described later in Section VII.

C. Model Selection (ML/DL)

This step focuses on the process of choosing between ML and DL models, considering factors such as the complexity of the dataset and the available computational resources.

Fig. 12 shows that 64% of researchers have built their threat model using different types of ML algorithms. While DL is considered to be more capable of finding complex patterns and more effective for large datasets, ML is more common in threat models. This is due to several reasons, including simplicity, interpretability, and fast training and deployment time.

GAN-based attacks can expose vulnerabilities of the threat model regardless of the model itself, whether it is ML or DL, unlike gradient-based attacks, which only target DL models. That is why most researchers focus on studying the effects of GAN-based attacks on ML-based threat models.

Fig. 13 illustrates the frequencies of various ML and DL models across the literature. When collecting and analyzing the

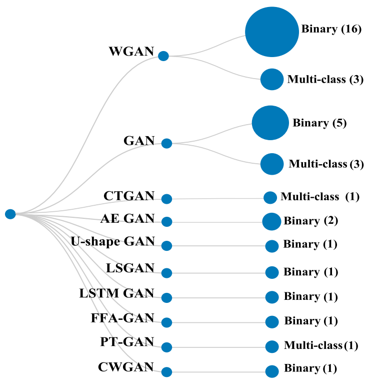

Fig. 16. Distribution of the type of classifier (threat model) and the GANs used for generating adversarial attacks.

Fig. 17. Distribution of the types of GAN used in each dataset for generating adversarial attacks.

studies, ANN, multilayer perception (MLP), and deep neural networks (DNN) were all classified as ANN. ANN, RF, and DT are the most studied models under GAN-based attacks.

In the last step of preparing the threat model, the process of training the model will take place, and after ensuring that the model achieves high-performance metrics, the process of testing the model will start. During testing, GAN-based adversarial scenarios will be simulated, as stated earlier.

VII. Adversarial Attack Generation

After establishing the threat model, the next step involves generating adversarial attacks that simulate potential real-world threats. This section discusses the methods and techniques used to generate such attacks in the last five years. The process of attacking the IDS aims to test its robustness under various conditions of GAN-based adversarial attacks. For generating GAN-based attacks, the steps shown in Fig. 14 should be considered.

A. GAN Architecture Selection

The first step in creating GAN-based adversarial attacks is to select the appropriate GAN architecture by determining the complexity and number of generators and discriminators of the GAN. Many common architectures exist, such as vanilla GANs, DCGAN, WGAN, CGANs, and others.

As depicted in Fig. 15, one of the most common GAN architectures used in GAN-based adversarial attacks is WGAN, and this is because it avoids the model collapse problem by using the Wasserstein loss function. Since 2022, researchers have started exploring different types of GANs, including BiGAN [9], cGAN [76], cWGAN [65], PTGAN [72], CTGAN [71], [56], and LSGAN [63]. The other category shown in Fig. 15 includes U-shape GAN [67], PT-GAN, FFA-GAN [64], LSTM GAN [68], and auto-encoder based GAN [66].

GAN-based adversarial attacks can be classified according to the number of generators and discriminators. By answering *RQ6*, we note that all existing GAN-based adversarial attacks used only one generator and one discriminator since it is easier to implement and understand. Moreover, using one generator and one discriminator avoids the additional complexity added to the training process when having multiple generators or multiple discriminators. Lastly, we can say that the GAN architecture used for generating attacks not only influences the effectiveness of these attacks in fooling the threat model but also affects the training process and the quality of the generated data. GANs' common problems, such as training stability and mode collapse, can be mitigated by carefully selecting the GAN architecture.

B. Defining the Game Model of the GAN

After determining the overall architecture of the GAN, the game model in which the generator and discriminator compete should be defined. According to [21], many game models can be used between the generator and discriminator. Upon studying the effect of *RQ5*, we found that all the existing studies use the basic min-max game, where the generator aims to minimize the likelihood of the discriminator correctly identifying adversarial samples, whereas the discriminator aims to maximize its accuracy by mimicking the threat model.

C. Training and Evaluating the GAN

After defining the game model of the GAN, the next step is to train it. The stability of the generator and discriminator is the most critical part of the training process to guarantee convergence, as stated in [77]. In GAN-based adversarial attacks, the generator is fed with the nonfunctional features of malicious traffic along with initial noise. The target of the generator is to fool the discriminator by combining the generator's output with the functional features of the original malicious traffic. On the other hand, the discriminator is trained to mimic the targeted model by taking the true labels of the adversarial malicious and normal records from the targeted model and using them for its training process.

The stability of the training process mainly depends on the previous steps (i.e., selected GAN architecture, loss function,

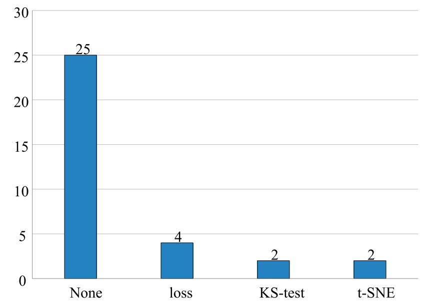

Fig. 18. Distribution of the metrics used to measure the quality of the generator's data.

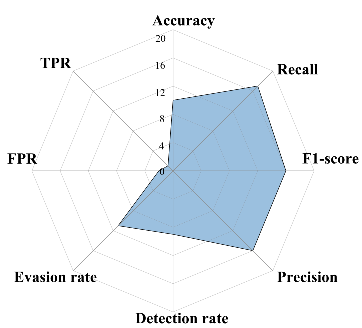

Fig. 19. Frequency of the metrics used to measure the severity of the adversarial attacks.

optimizer, game model, and many other parameters). Additionally, data quality and diversity are considered to be of utmost importance in achieving stable training.

To ensure the training stability for the discriminator part in trying to mimic the targeted model, the effects of *RQ2* and *RQ4* are studied as shown in Fig. 16. It can be noticed that some types of GANs, such as WGAN and GAN, are preferable to use with binary classifiers rather than multiclass classifiers.

As mentioned before, using binary classifiers makes it easier for the generator to generate adversarial samples, which affects the complexity and stability of the GAN's training process. This is why the basic types of GANs, such as WGAN and vanilla GAN, tend to focus more on binary classifiers, while more sophisticated GANs, such as CTGAN and PT-GAN, prove their efficiency in multiclassification models.

Because data quality and diversity also affect the training process of GANs, the effects of *RQ3* and *RQ4*, shown in Fig. 17, are studied. Once again, WGAN and GAN are the most used types of GANs across different datasets, meaning that simple GAN types can effectively capture the nature of the data. Nevertheless, this might not be the case when talking about newer datasets. When considering private datasets "their own" more advanced GANs were used, such as LSGAN and CWGAN.

Measuring the system's performance and the quality of the data generated by the system is extremely important. However, as shown in Fig. 18, most existing studies did not consider this crucial point. This might be because only the nonfunctional features of the malicious data are being modified, and the quality of the generated data was not considered. However, it is essential to study how close these adversarial samples are to

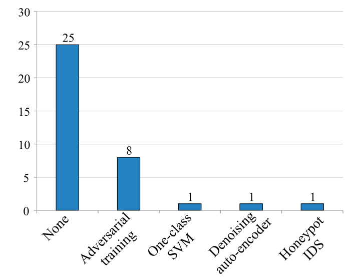

Fig. 20. Frequency of the defense mechanisms in the literature.

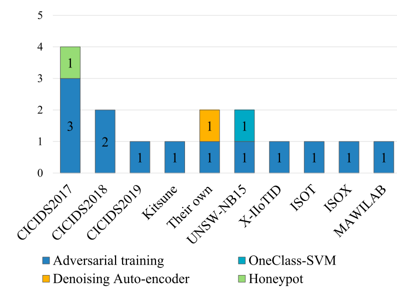

Fig. 21. Frequency of the types of defense mechanisms applied to each dataset.

the distribution of normal data and how similar they are, either by using the loss of the GAN [45], [63], [65] or by using one of the other metrics such as KS-test [37], Hellinger distance [37], t-SNE [54], [67], or Jaccard index [5].

In the last step, the generator will be deployed to generate adversarial samples that evade the target model (i.e., the IDS). The severity of these attacks will be further discussed in the next section. The main goal of this step is to identify the vulnerabilities of the target model and how it will respond to the adversarial samples, which in turn helps us build an effective defense mechanism.

VIII. Severity Assessment and Defense Mechanism

Once the adversarial attacks have been generated, it is crucial to evaluate their impact on the threat model. By understanding how the threat model is affected under attack, it becomes easier to understand the system's vulnerabilities and propose a defense mechanism accordingly.

To explore how GAN-based attacks affect the threat model in the literature, we first look into the effects of *RQ8*, which is concerned with studying all metrics used to measure the impact of GAN-based adversarial attacks. As represented in Fig. 19, most researchers use the traditional metrics to measure the performance of the IDS under attack, and few of them use the metrics dedicated to measuring the evasiveness of the adversarial attacks, such as the detection rate [39], [43], [44], [49], [52], [54], [56], [69], [78], [79] and evasion rate [36], [38], [39], [43], [44], [46], [52], [54], [56], [65], [79], [80], [81], [82]. It is worth noting that the most used metrics for anomaly detection are the recall, precision, and F1-score; since such metrics consider the false positive rates, which is what matters in adversarial attack. Moreover, the existing metrics

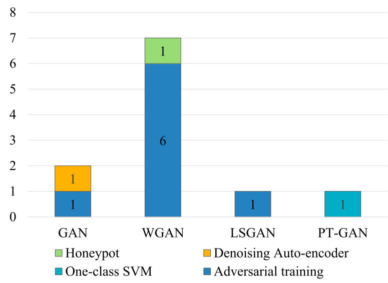

Fig. 22. Frequency of the types of defense mechanisms applied to each attack.

only measure the evasiveness rather than the maliciousness of these attacks.

After evaluating the severity of GAN-based adversarial attacks, it becomes clear to understand the system's overall security threats and highlight the necessary steps for enhancing the robustness of IDS models. Upon studying the effects of *RQ9*, we found that there are many ways to defend IDSs against GAN-based adversarial attacks. As shown in Fig. 20, most researchers focused on three main aspects of detecting GAN-based adversarial attacks: Adversarial training [46], [58], [63], [69], [78], [83], secondary classifier [72], and input preprocessing such as denoising auto-encoder [45], [55]. It is to be noted that most existing studies do not propose defense mechanisms, despite their importance.

Regardless of the type of GAN-based adversarial attacks used in the literature, almost all of them depend on adversarial training. Hence, the effects of the used dataset and the selected defense mechanism should be studied. Fig. 21 represents the effects of *RQ3* and *RQ9*, which reflect the relationship between the dataset and the type of defense mechanism used. It can be noticed that WGAN has been studied the most in almost all defense mechanisms, and this is because it is considered to be the most used GAN-based adversarial attack. Meanwhile, the dataset that has been studied the most in terms of defense techniques is the CICIDS2017 dataset, as it is the most used dataset for training threat models.

The GAN-based adversarial attacks can influence the nature of the adversarial samples. As a result, the effects of *RQ4* and *RQ9* have been analyzed to explore the characteristics of adversarial examples produced by a particular GAN type and the choice of defense strategy. It can be noted from Fig. 22 that most defense strategies were applied to WGAN because of its convergence and stability.

IX. Discussion and Future Directions

In this section, the main findings of our systematic review are discussed and analyzed to provide future directions in the field of GAN-based adversarial attacks against IDSs.

A. Threat Model

1) Dataset Selection: In the context of building a threat model, the first step is to select an appropriate dataset, as the data quality and diversity affect the performance of IDSs, as stated in [84]. It has been noticed that most datasets used are considered outdated, as the most recent one that has been studied and analyzed against GAN-based attacks was CICIDS2019. The CICIDS2019 dataset and similar datasets simulate traditional network traffic, and it depends on 4G or earlier networks, where the traffic is more static and homogeneous. Nonetheless, most current networks use 5G technologies in different fields. As a result, most recent IDSs use 5G network traffic [85], [86]. Thus, it is essential to study the impact of GAN-based adversarial attacks on IDSs built upon more up-to-date datasets.

Moreover, to make the GAN-based attacks independent of the type of dataset used, GANs can be used at the packet level rather than the feature level. This might be more challenging because attacks can be generated using a sequence of packets.

2) Preprocessing: Preprocessing is an essential step in developing any model, and two key aspects affect GAN attack severity: 1) label encoding; and 2) feature selection. The first important consideration is how to encode the label, which can be done either as a binary or a multiclass format. From the adversary perspective, it is easier for attackers to generate adversarial attacks targeting binary classifiers than multiclass ones, as shown in the following section. On the other hand, from the defender's perspective, using multiclass classifiers is advantageous, as it allows for more tailored responses to each type of attack.

In the context of feature selection, the impact of GAN-based attacks should be studied when feature selection or dimensionality reduction is considered, since the generator modifies only the nonfunctional features of the traffic, and these features may not be considered when applying feature selection or dimensionality reduction algorithms.

3) Model Selection: One of the most critical factors that helped in the development of IDSs is the abundance of big data. To process this amount of data, increase the efficiency of IDSs, and discover hidden patterns, IDSs use DL. However, as discussed before, the most commonly used threat models were built upon ML. Hence, the ability of GAN-based attacks to mislead ML models may not be applied to DL models.

The adversarial attacks can happen either at the training step (poisoning attack) or the testing step (evasion attack). GAN-based attacks are considered black-box attacks, which means they are one type of evasion attack. Nevertheless, it can be used as a poisoning attack by injecting adversarial samples into the training step. Still, the attacker's goal here is to postpone the deployment of the IDS, since a model with poor performance at the training step will not be deployed, and this can benefit the attacker in taking advantage of the vulnerabilities that exist in the current IDS.

B. Adversarial Attack Generation

1) GAN Architecture Selection: As mentioned before, GANs are powerful tools that can be used to generate adversarial attacks. Referring to Fig. 14, determining the GAN's architecture is the first step to consider when employing GAN-based attacks. Modifying the architecture and loss functions can help improve GANs. Vanilla GANs and WGANs are the most common methods among researchers in the field of attacking IDSs, and the reason behind using simpler GAN architectures lies in the balance between efficiency and effectiveness. Simpler GANs, such as vanilla GANs and WGAN, are straightforward to train and require fewer computational resources, which aligns with the goal of crafting adversarial samples with minimal overhead. Additionally, they have demonstrated an adequate ability to generate traffic that causes misclassification, thus appearing to successfully bypass IDS models. However, we argue that bypassing the decision boundary alone is not a sufficient measure of success. The quality and stealthiness of the generated adversarial traffic are crucial for real-world scenarios. Low-quality adversarial samples may raise suspicion or be easily detected by anomaly-based systems or even manual inspection.

Moreover, the use of simpler GAN-based attacks has shaped the design of defense strategies. Many existing defense mechanisms are designed to counter basic adversarial perturbations generated by common GANs. As a result, these defenses might not generalize against more sophisticated adversarial attacks when using advanced GAN architectures. Many other types of GANs prove their efficiency in cybersecurity, such as the InfoGAN and CycleGAN, as experimentally proven in [87], [88]. These types of GANs can also be applied and studied in the context of GAN-based attacks. GAN architectures can also refer to the number of generators and discriminators used. Utilizing GANs with multiple generators has been used by many researchers in the field of IDSs for data augmentation [89] and anomaly detection [90]. Hence, it is worth studying how multigenerator GANs can influence the generation of adversarial attacks. Furthermore, a promising future direction for adversarial sample generation lies in the use of model fusion, which solves GANs' instability and mode collapse problems. In model fusion, multiple generative or predictive models are combined to enhance the diversity and effectiveness of adversarial attacks. By leveraging the strengths of different architectures.

2) Defining the Game Model of the GAN: One of the key factors that has a direct impact on the quality of the data produced by GANs is the game model used to train the generator and discriminator. Zhang et al. [91] showed that using different types of models enhances the diversity of the newly generated data, and this could be applied in the field of GAN-based adversarial attacks. In other words, using other game models in GAN-based attacks may generate a large diversity of attacks, making it hard to detect the adversarial samples. Moreover, using game-theoretic models is not new in the field of generating adversarial samples; Brückner et al. [92] used a Stackelberg game to generate spam emails. Hence, using GANs based on different game models may not only enhance the quality of the newly generated adversary data but also speed up the process of generating adversary samples.

3) Evaluating the GAN: As the primary goal for GANs is to produce realistic data similar to the original one, it is important to verify the quality and authenticity of the generated data using different types of evaluation metrics. In the context of GAN-based attacks, crossing the decision boundary is very important for the success of the attack, but measuring how close the adversary samples are to the authenticated samples is just as important. Therefore, evaluation metrics, such as the IS and FID, are essential to confirm that the GANs are fulfilling their intended purpose.

C. Severity Assessment and Defense Mechanisms

To say that GAN-based adversarial attacks are successful and achieve the attacker's goals, a severity assessment should be conducted to measure the effectiveness of these attacks. The use of normal metrics such as accuracy, recall, precision, F1-measure, FPR, and TPR is often insufficient because IDSs might still have a high overall accuracy but be highly vulnerable to adversarial attacks on specific inputs. Therefore, other metrics, such as evasion rate and detection rate, have been used widely across the literature. However, crossing the decision boundary is not enough for adversarial attacks to be successful; IDSs should confidently misclassify the adversarial samples. Measuring severity requires new metrics that consider these broader impacts. In other words, new metrics must be used to measure the maliciousness of the adversary samples.

After discovering the vulnerabilities of existing threat models, decision-makers should take action and develop defense mechanisms to increase the robustness of these threat models. Nonetheless, defending against GAN-based adversarial attacks is more difficult than generating the attacks. Most research papers merely point to the existence of vulnerabilities in IDSs. The difficulty of defending against GAN-based adversarial attacks derives from the necessity of understanding the system's weaknesses and understanding the attack itself so that decision-makers can take the required actions.

Among the few proposed defense strategies, adversarial training is the most common approach. However, adversarial training confuses the model because the model is forced to learn a more complex decision boundary to accommodate adversarial examples, which can degrade its performance. Moreover, creating more complex boundaries can lead to the overfitting problem. Apart from these issues, the training time increases drastically depending on the classifier used in the IDS, and since adversarial training is vulnerable to adaptive attacks, the process of retraining the model happens frequently. Another common defense mechanism in the literature is the use of an ensemble approach that incorporates various statistical methods, such as a multiarmed bandit strategy [75], to dynamically select and combine models, thereby enhancing robustness against transferable adversarial attacks.

Adversarial attacks can be seen as added noise to the network traffic. Therefore, some researchers tend to use denoising auto-encoders and randomized smoothing. However, auto-encoders are also subject to adversarial attacks, and building an auto-encoder can increase the training time, as stated in [73]. This also applies to the use of secondary classifiers. Thus, there is a need for a new way to defend threat models, starting from the preprocessing techniques to the process of deploying and evaluating the model. The preprocessing techniques include limiting the number of features an attacker can manipulate.

Regarding the model selection, ensemble models can be more robust than other models because GAN-based adversarial samples usually try to mimic the IDS. If there is more than one model, it is hard for the discriminator to accurately mimic all those models. This, in turn, affects the overall quality of the adversarial sample, reducing its possibility of deceiving the IDS. A game-theoretic defense framework similar to the one used in SGAN could significantly enhance the robustness of IDSs. In such a framework, the interaction between attackers and defenders is modeled as a strategic game, allowing for the anticipation of adversarial behavior and the optimization of defensive strategies. This optimization can be achieved through various techniques, such as ANN pruning, dynamic feature selection, and other potential actions.

Another important future direction involves addressing the black-box nature of DL/ML models, which remains a fundamental vulnerability exploited by adversarial attacks. Incorporating explainability and interpretability techniques could provide valuable insights into how and why models are susceptible to such manipulations. Despite being a growing area, explainable AI (XAI) remains largely unexplored within GAN-based adversarial attack and defense. Bridging this gap could significantly enhance our understanding of adversarial behavior and lead to more transparent and robust IDS.

Federated learning (FL) represents a promising future direction for enhancing robustness against adversarial attacks due to its distributed nature. By combining FL with various defense mechanisms, it is possible to mitigate the effects of malicious clients or adversarially crafted updates. This can be achieved by identifying and down-weighting or excluding abnormal contributions to the model during the aggregation process. Although FL is not entirely immune to sophisticated threats [93], it provides a flexible and privacy-preserving framework. This allows for the integration of diverse defense strategies across distributed clients, enabling more resilient and adaptive protection mechanisms.

X. Experiments and Evaluation

This section presents the conducted experiments and procedures to validate the key points discussed in Section IX. The experiments are designed to assess the statements mentioned earlier, offering empirical evidence to support our claims.[1]

Our approach is built upon utilizing the most common methods in the literature, providing a framework to evaluate the impact of GAN-based attacks on the 5G dataset created in [94]. By testing the impact of such attacks under various conditions, we aim to strengthen the following assertions.

1) GAN-based attacks built upon more up-to-date datasets have a significantly higher impact on current IDSs.
2) The effectiveness of GAN-based attacks against binary and multiclass models can vary, and this should be investigated in terms of: 1) the most and the least affected class; and 2) the diversity of the GAN's output.
3) Feature selection techniques play a significant role in limiting the number of features an attacker can manipulate, thereby reducing the effectiveness of adversarial attacks.

We conducted our experiments on a Windows machine with an Intel Core i7 processor and 16 GB RAM. We used Python 3.8 to implement our experiments. Aligned with the most commonly used model in the literature, we designed our experiments to attack ANN-based IDSs. To study the impact of adversarial attacks on IDSs, we utilized two versions of the ANN with the same architecture, a binary classification model, and a multiclass classification model. We used GAN and vanilla WGAN [29], the two most common GANs, to generate adversarial attacks. Furthermore, we have incorporated three advanced GAN architectures: 1) BiGAN; 2) SGAN; and 3) CycleGAN. They were systematically used to generate adversarial samples aimed at deceiving the two versions of the ANN model. Therefore, we conducted the following experiments: 1) WGAN attacking a binary IDS; 2) WGAN attacking a multiclass IDS; 3) GAN attacking a binary IDS; 4) GAN attacking a multiclass IDS; 5) BiGAN attacking a binary IDS; 6) BiGAN attacking a multiclass IDS; 4) SGAN attacking a multiclass IDS; 8) CycleGAN attacking a binary IDS; and 9) CycleGAN attacking a multiclass IDS.

We repeated the same experiments with feature selection to study the impact of feature selection on GAN-based adversarial attacks. The experiments were replicated on the CICIoT2023 dataset to ensure robustness and generalizability. We observed results that were highly consistent with those reported for 5G-NIDD. The following sections provide further details on each stage of conducting the experiments.

A. Threat Model Development

1) Dataset Selection: As mentioned before, there is a need to study the impact of GAN-based adversarial attacks on IDSs built upon 5G network technologies. The 5G network intrusion detection dataset (5G-NIDD) [94] offers comprehensive features that capture various network traffic under real settings. This dataset consists of 112 features and 1 048 567 records with eight different types of attacks, which include 1) SYNScan (20 043); 2) SYNFlood (4792); 3) UDPScan (445 708); 4) TCPConnectScan (20 052); 5) UDPFlood (15 906); 6) HTTPFlood (76 121); 7) SlowrateDoS (36 092); and 8) ICMPFlood (1155). The 5G-NIDD dataset has 619 869 malicious records and 428 503 benign records. To ensure the robustness and generalizability of our findings, we extended our evaluation to a second dataset, the CICIoT2023 dataset [95], which is a recently introduced benchmark reflecting contemporary IoT network traffic. For consistency, across experiments, we restricted our analysis to the attack types represented in the 5G-NIDD dataset.

2) Preprocessing: The preprocessing step involves preparing the dataset for training and testing by cleaning the raw data and transforming it into a well-structured format. This step is crucial to build an accurate model.

The preprocessing consists of four main stages: Feature selection, data cleaning, feature encoding, and normalization. In the feature selection stage, all features with the same

[1]Codes are available at: https://github.com/Yasmeen1710/Exploiting-GANs-Against-IDSs-Code

TABLE XI
MODEL ARCHITECTURE

Layer	Output Shape	Activation Function
1	(128,)	ReLU
2	(64,)	ReLU
3	(64,)	ReLU
4	(8,)	ReLU
5 (ANN-1, 3, 5)	(1,)	Sigmoid
5 (ANN-2, 4, 6)	(9,)	Softmax

value across all records are eliminated. Internet protocol (IP) addresses and port numbers are also eliminated, so the IDS can focus on the network traffic's behavioral patterns. In the data cleaning phase, all null values are either dropped or replaced by zero or the median value, depending on the feature, in both datasets. Additionally, all categorical features in the 5G-NIDD dataset ("Proto", "sDSb", "Cause", "State", "Label", "Attack Type", "Attack Tool") are encoded using the label encoder technique.

After the preprocessing step, there were 31 features and 1 048 372 records in the first version, referred to as 5G-NIDD-V1. It is to be noted that two additional versions of the datasets have been created from 5G-NIDD to achieve the objectives of these experiments. The second version, 5G-NIDD-V2, contains only the top ten features of the 31 features. In contrast, the third version, 5G-NIDD-V3, contains the functional and categorical features of the denial of service (DoS) attack. According to [54], the functional features of DoS attacks are the active-mean, flow duration, mean packet size, mean time between two packets, and max-idle time.

3) Model Selection: Six ANN models are utilized in the conducted experiments for each dataset (5G-NIDD and CICIoT2023). ANN-1 and ANN-2 refer to the binary and multiclass models, respectively. ANN-3 and ANN-4 refer to the binary and multiclass models, respectively, with the top-ten features. ANN-5 and ANN-6 refer to the binary and multiclass models, respectively, using only the functional features. To preserve consistency, all models share the same architecture except for the output and input layers. Table XI summarizes the ANN architecture.

The stochastic gradient descent (SGD) optimizer is used for all models. ANN-1, 3, 5 use the binary cross-entropy loss function, whereas ANN-2, 4, 6 use the sparse categorical cross-entropy. All models are trained on 67% of the data and tested on the remaining 33%.

4) Model Evaluation: Evaluating the model is essential to guarantee its effectiveness. In this section, we analyze the models based on accuracy, precision, recall, and F1-measure, where TP, TN, FP, and FN denote the true positives, true negatives, false positives, and false negatives, respectively. $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$, $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$, $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$, and $\text{F1-Score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$.

These metrics provide insights into how well the models can distinguish between benign and malicious traffic. Table XII shows these metrics for ANN-1, ANN-3, ANN-5, ANN-2, ANN-4, and ANN-6, respectively.

Note that the performance of IDSs is almost the same in ANN-1 and ANN-3. ANN-5 for both datasets performed the worst when using only the functional features of the DoS attack. Nonetheless, it is the most robust, as shown in the following section.

ANN-2 and ANN-4 are the most effective multiclass models. ANN-6 needs more consideration, as it fails to classify three attacks. This means the functional features of DoS are not sufficient to build an effective IDS. More features should be included in ANN-5 and ANN-6 in an optimized manner to enhance overall performance by adding the least number of nonfunctional features. Similar results were obtained when training the model on the CICIoT2023 dataset.

B. Adversarial Attack Generation

GAN-based adversarial attacks significantly impact IDSs, especially when built upon more up-to-date datasets. Therefore, the two most common GAN architectures, in addition to BiGAN and SGAN, are used in this experiment. Following the steps of IDSGAN [5], the features are divided into three main categories: 1) intrinsic; 2) functional; and 3) nonfunctional. Intrinsic features are the features found in the packet header. Nonfunctional features are the features that can be manipulated by the attacker to mislead IDSs after excluding the categorical ones. Whereas the functional features are the features that directly contribute to harming the targeted service or system. 5G-NIDD-V1 contains 23 nonfunctional features and 5G-NIDD-V2 contains seven. This means that the latent vector (noise vector) for the generator is 23 for 5G-NIDD-V1 and seven for 5G-NIDD-V2.

1) GAN Architectures: As the generator's primary role is to generate adversarial traffic as close to the normal traffic data as possible, the architectures of the generators begin with an input layer that matches the dimensionality of the noise vector. The noise vector can vary in size depending on the number of nonfunctional features the targeted model has. It then passes through three dense layers with ReLU activation functions, with 128, 64, and 32 neurons, respectively. The last layer of the generators is the output layer, which uses the tanh as an activation function. The generator's output is then added to the nonfunctional features of the malicious records. The generator is trained to produce adversarial traffic so that the discriminators cannot distinguish the adversarial samples from normal data.

On the other hand, the discriminators are designed to classify the traffic by mimicking the IDS. The labels for both the generated adversarial samples and the normal samples are obtained by querying the IDS. The discriminator architecture starts with an input layer and passes through two dense layers with 128 and 64 neurons, respectively, each with a ReLU activation function.

We employ four versions of GAN, WGAN, BiGAN, CycleGAN, and two versions of SGAN as a foundation for generating adversarial samples. GAN-V1, WGAN-V1, BiGAN-V1, and CycleGAN-V1 are all designed to attack ANN-1. GAN-V2, WGAN-V2, BiGAN-V2, CycleGAN-V2, and SGAN-V1 are developed to attack ANN-2. ANN-3 was attacked by GAN-V3, WGAN-V3, BiGAN-V3, and CycleGAN-V3. GAN-V4,

Algorithm 1: Training GAN-Based Adversarial Attacks.

Require: Normal and malicious traffic records $S_{\text{normal}}, S_{\text{attack}}$; the noise z for the adversarial generation; the trained classifier IDS the generator G and the critic D

for epoch in epochs **do**
  Sample a batch of normal data x_{normal} from S_{normal}
  Sample a batch of malicious data $x_{\text{malicious}}$ from S_{attack}
  for k steps of the critic training **do** ▷ k is 1 if Vanilla GAN
    Generate fake data x_{fake} using $G(x_{\text{malicious}}, z)$
    Compute labels y_{normal} using IDS on x_{normal}
    Compute labels y_{fake} using IDS on x_{fake}
    Train the critic D on x_{normal} with both y_{normal}
    Train the critic D on x_{fake} with y_{fake}
  end for
  Train the generator G with the critic D
end for

Algorithm 2: Training SGAN-Based Adversarial Attacks.

Require: Normal and malicious traffic records $S_{\text{normal}}, S_{\text{attack}}$; the noise z for the adversarial generation; the trained classifier IDS; the generator G (a list of generators); the critic D; number of critic steps k

for epoch in epochs **do**
  Sample a batch of normal data x_{normal} from S_{normal}
  Compute labels y_{normal} using IDS on x_{normal}
  for G_c in G **do** ▷ Each generator G_c targets class c
    for $i = 1$ to k **do** ▷ k steps of critic training
      Sample class-specific data $x^{(c)}_{\text{malicious}}$ from S_{attack}
      Generate $x^{(c)}_{\text{fake}}$ using $G_c(x^{(c)}_{\text{malicious}}, z)$
      Compute labels $y^{(c)}_{\text{fake}}$ using IDS on $x^{(c)}_{\text{fake}}$
      Train the critic D on x_{normal} with y_{normal}
      Train the critic D on $x^{(c)}_{\text{fake}}$ with $y^{(c)}_{\text{fake}}$
    end for
    Train the generator G_c with the critic D
  end for
end for

WGAN-V4, BiGAN-V4, CycleGAN-V4 and SGAN-V2 are implemented to attack ANN-4. All versions share the same architecture except for the size of the noise vector for the generator and the model that the discriminator is trying to mimic.

In vanilla GAN, both the generator and discriminator share the same architecture as WGAN. However, the output layer in the discriminator has a sigmoid activation function, and the loss function for the vanilla GAN is binary cross-entropy instead of the Wasserstein loss function. The training algorithm of the vanilla GAN and WGAN is represented in Algorithm 1. In SGAN, each class of malicious traffic is handled by a dedicated generator, enabling class-specific adversarial example generation, which in turn increases the adversarial sample diversity. During training, as presented in Algorithm 2, each generator is paired with a shared critic and trained independently in a loop. This design allows SGAN to preserve class semantics better while crafting more targeted and diverse attacks. BiGAN was also utilized to generate adversarial samples. BiGAN extends the standard GAN framework by introducing an encoder network that learns to map real data back to the latent space as it jointly trains an encoder, generator, and discriminator to align real data, latent code pairs with generated ones, as presented in Algorithm 3. CycleGAN employs two generators that learn bidirectional mappings between two domains, supported by discriminators and the cycle-consistency constraint, as represented in Algorithm 4. In the context of GAN-based adversarial attack, this bidirectional mapping enables more advanced adversarial examples, as the generator can exploit latent representations learned from normal traffic.

Algorithm 3: Training BiGAN-Based Adversarial Attacks.

Require: Normal and malicious traffic records $S_{\text{normal}}, S_{\text{attack}}$; noise z; trained classifier IDS; encoder E; generator G; discriminator D

for epoch in epochs **do**
  Sample a batch of normal data x_{normal} from S_{normal}
  Sample a batch of malicious data $x_{\text{malicious}}$ from S_{attack}
  Encode x_{normal} to obtain latent codes: z_{enc} using $E(x_{\text{normal}})$
  Generate adversarial samples: x_{fake} using $G(x_{\text{malicious}}, z)$
  Compute labels y_{normal} using IDS on x_{normal}
  Compute labels y_{fake} using IDS on x_{fake}
  Train the critic D on x_{normal} and z_{enc} with y_{normal}
  Train the critic D on x_{fake} and z with y_{fake}
  train G and E jointly to fool D and mislead the IDS
end for

Algorithm 4: Training CycleGAN-Based Adversarial Attacks.

Require: Normal and malicious traffic records $S_{\text{normal}}, S_{\text{attack}}$; the trained classifier IDS; the generators $G_{M\to N}$ and $G_{N\to M}$; the discriminator D

for epoch in epochs **do**
  Sample a batch of normal data x_{normal} from S_{normal}
  Sample a batch of malicious data $x_{\text{malicious}}$ from S_{attack}
  Generate fake normal samples: $x_{\text{fake}} = G_{M\to N}(x_{\text{malicious}})$
  Generate cycle-reconstructed malicious: $\hat{x}_{\text{malicious}} = G_{N\to M}(x_{\text{fake}})$
  Compute adversarial labels y_{fake} using IDS on x_{fake}
  Train discriminator D to distinguish real vs. fake in their respective domains
  Train generators $G_{M\to N}$ and $G_{N\to M}$ to fool D and to Minimize cycle consistency losses
end for

TABLE XII
MODELS' PERFORMANCE

Model Name	Accuracy	Precision	Recall	F1-Measure
ANN-1	99.9%	99.9%	99.8%	99.9%
ANN-3	99.9%	99.9%	99.9%	99.9%
ANN-5	72.9%	68.6%	99.9%	81.3%
ANN-2	99%	98%	96%	97%
ANN-4	98%	93%	93%	93%
ANN-6	96%	70%	66%	62%

TABLE XIII
ANN-1 AND ANN-3 UNDER GAN-BASED ATTACKS

Model Name	GAN Architecture	Accuracy	Precision	Recall	F1-Measure	FID	HD
ANN-1	WGAN-V1	41%	20%	50%	29%	2.3	0.55
ANN-1	GAN-V1	41%	20%	50%	29%	31.3	0.73
ANN-1	BiGAN-V1	41%	20%	50%	29%	20.9	0.72
ANN-1	CycleGAN-V1	41%	20%	50%	29%	3.9	0.61
ANN-3	WGAN-V3	41%	20%	50%	29%	0.46	0.36
ANN-3	GAN-V3	41%	20%	50%	29%	7.7	0.56
ANN-3	BiGAN-V3	41%	20%	50%	29%	0.46	0.55
ANN-3	CycleGAN-V3	41%	20%	50%	29%	4.2	0.51

C. Severity Assessment

In this section, the severity of GAN-based adversarial attacks is studied and analyzed based on the accuracy, recall, precision, and F1-score, which measure the performance of the IDSs under GAN-based attacks. In addition, FID and HD are used to measure the quality of the generated adversarial samples and how well adversarial attacks inject falsified data into the system. The attack success rate is also used to study the severity of GAN-based attacks. By analyzing these metrics, we can

TABLE XIV
ANN-2 AND ANN-4 UNDER GAN-BASED ATTACKS

Model Name	GAN Architecture	Acc- uracy	Prec- ision	Rec- all	F1-Measure	FID	HD	ASR
ANN-2	WGAN-V2	43%	22%	16%	14%	4.1	0.57	92%
ANN-2	GAN-V2	45%	21%	15%	14%	17.9	0.75	99%
ANN-2	BiGAN-V2	41%	5%	11%	6%	31.7	0.73	100%
ANN-2	SGAN-V1	47%	50%	29%	29%	0.87	0.53	84%
ANN-2	CycleGAN-V2	41%	38%	11%	6%	18.1	0.73	92%
ANN-4	WGAN-V4	47%	45%	19%	18%	1.23	0.39	78%
ANN-4	GAN-V4	73%	29%	19%	18%	11.4	0.57	90%
ANN-4	BiGAN-V4	41%	5%	11%	6%	5.13	0.57	100%
ANN-4	SGAN-V2	59%	45%	30%	31%	1.02	0.32	79%
ANN-4	CycleGAN-V4	41%	18%	11%	6%	3.4	0.56	100%

comprehensively assess and confirm the assertions mentioned earlier.

1) Attacking Binary Models: ANN-1 and ANN-3 are attacked by WGAN-V1, WGAN-V3, GAN-V1, GAN-V3, BiGAN-V1, BiGAN-V3, CycleGAN-V1, and CycleGAN-V3. The performance of ANN-1 under WGAN, GAN, BiGAN, and CycleGAN attacks is summarized in Table XIII.

When attacking a binary classifier, all types of GANs impact the IDS's performance similarly, reducing their accuracy from 99% to almost 41%. However, when evaluating the FID score and HD, which reflect the quality and stealthiness of the attacks, WGAN appears to generate the most realistic adversarial examples, as it achieves the lowest HD and FID compared with the normal class. It is worth noting that the attack success rate in all types of GANs was 100%.

Feature selection may seem like a viable strategy to mitigate the impact of adversarial attacks. However, relying on a simple approach, such as selecting the top ten features, can lead to the same poor performance under adversarial attacks. This is because each selected feature plays a significant role in the final decision, increasing the model's vulnerability to adversarial manipulation. On the other hand, ANN-5 maintains its performance under both attacks as it does not use any nonfunctional features.

2) Attacking Multiclass Models: Attacking multiclass models is considered to be more complicated than attacking binary classifiers, as there are complex decision boundaries. The results confirm this assertion, as shown in Table XIV.

Among the evaluated GAN architectures, BiGAN and CycleGAN exhibited the most severe impact on the IDS, as indicated by the lowest precision, recall, and F1-scores across both ANN-2 and ANN-4 models. However, SGAN generated the most realistic adversarial samples, achieving the lowest HD and FID values, suggesting its outputs closely resemble normal traffic. In contrast, the standard GAN architecture consistently demonstrated the poorest performance, offering neither strong evasion capabilities nor realistic samples. While WGAN affected the IDS performance more noticeably than SGAN in terms of performance degradation, SGAN's adversarial examples were more stealthy and indistinguishable from legitimate traffic.

The ANN-4 model, which incorporates feature selection, shows a notable variation in its robustness against different GAN-based attacks. BiGAN-V4 caused the most severe performance degradation, with the lowest precision, indicating its strong ability to generate adversarial samples that mislead the IDS even with selected features. On the other hand, SGAN-V2 produced the most realistic adversarial traffic, achieving the lowest FID (1.02) and HD (0.32) values, highlighting its capability to closely mimic normal traffic patterns. Interestingly, while WGAN-V4 led to greater performance degradation than SGAN-V2, SGAN-V2 remained more stealthy, reinforcing that feature selection improves IDS performance but also challenges its ability to detect well-crafted adversarial examples. Notably, standard GAN-V4 underperformed across all metrics, suggesting it is the least effective in both deceiving the IDS and generating realistic traffic under feature selection. Similar to BiGAN-based adversarial attacks, CycleGAN-based adversarial attacks have a substantial impact on IDS performance, affecting metrics such as accuracy, precision, recall, and F1-measure. However, CycleGAN attacks are notably more stealthy than those generated by BiGAN, making them a more challenging threat.

The severity assessments conducted in these experiments confirmed the first assertion, i.e., IDSs based on up-to-date datasets are also subjected to adversarial attacks. Regarding the second assertion, it has been shown that multiclass adversarial attacks are more complex and need more consideration than binary attacks. Finally, the third assertion is that while feature selection can reduce the attack surface by limiting the number of manipulable features, it must be applied with caution. The results from ANN-4, which employs feature selection, reveal that common approaches, such as selecting the top ten features, may inadvertently weaken the IDS. In particular, models using such selections showed severe performance degradation under BiGAN-based and CycleGAN-based attacks and struggled to detect highly realistic adversarial samples generated by SGAN. This highlights the need for a more strategic and optimized feature selection to maintain IDS robustness against advanced GAN-based threats. All assertions were confirmed when the experiments were replicated on the CICIoT2023 dataset, with only minor variations in the numerical results.

XI. CONCLUSION

In this article, we systematically and comprehensively reviewed the literature related to GAN-based adversarial attacks on IDSs and their defense mechanisms. A deep understanding of the current trends in exploiting GAN-based adversarial attacks was established, driven by an extensive meta-analysis conducted based on critical review questions. Our meta-analysis revealed the limitations of state-of-the-art GAN-based attacks on IDSs in terms of their threat models, attack generation, defense mechanisms, and severity assessment methods. Motivated by these limitations, we employed three innovative GAN architectures, BiGAN, SGAN, and CycleGAN, along with the state-of-the-art GANs, including GAN and WGAN, to generate and evaluate these attacks on IDSs trained on a 5G dataset and on an IoT dataset, considering 32 representative use cases. We defined our use cases assuming different classifiers (binary, multiclass) with and without feature selection being enabled. Among the evaluated GANs, BiGAN produced the most severe attacks, and SGAN generated the most realistic adversarial samples. The studied use cases demonstrated: 1) the need for a more comprehensive severity assessment of GAN-based attacks, specifically when targeting multiclass IDSs; and 2) the significant impact of optimal selection of nonfunctional features on enhancing IDS resiliency.

REFERENCES

[1] I. Goodfellow et al., "Generative adversarial networks," *Adv. Neural Inf. Process. Syst.*, vol. 27, pp. 139–144, 2014.

[2] W. Xia, Y. Zhang, Y. Yang, J.-H. Xue, B. Zhou, and M.-H. Yang, "GAN inversion: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3121–3138, Mar. 2022.

[3] N. E. Khalifa, M. Loey, and S. Mirjalili, "A comprehensive survey of recent trends in deep learning for digital images augmentation," *Artif. Intell. Rev.*, vol. 55, no. 3, pp. 2351–2377, 2022.

[4] Y. Cui, Z. Liu, and S. Lian, "A survey on unsupervised anomaly detection algorithms for industrial images," *IEEE Access*, vol. 11, pp. 55297–55315, 2023.

[5] Z. Liu, J. Hu, Y. Liu, K. Roy, X. Yuan, and J. Xu, "Anomaly-based intrusion on IoT networks using AIGAN-A generative adversarial network," *IEEE Access*, vol. 11, pp. 91116–91132, 2023.

[6] J. Fu, L. Wang, J. Ke, K. Yang, and R. Yu, "GANAD: A GAN-based method for network anomaly detection," *World Wide Web*, vol. 26, no. 5, pp. 2727–2748, 2023.

[7] H. Ding, Y. Sun, N. Huang, Z. Shen, and X. Cui, "TMG-GAN: Generative adversarial networks-based imbalanced learning for network intrusion detection," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 1156–1167, 2023.

[8] V. Kumar and D. Sinha, "Synthetic attack data generation model applying generative adversarial network for intrusion detection," *Comput. Sci.*, vol. 125, 2023, Art. no. 103054.

[9] N. AbdAL-Gawad, A. Sajun, Y. Kaddoura, I. A. Zualkernan, and F. Aloul, "Generative deep learning to detect cyberattacks for the IoT-23 dataset," *IEEE Access*, vol. 10, pp. 6430–6441, 2021.

[10] H. Navidan et al., "Generative adversarial networks (GANs) in networking: A comprehensive survey & evaluation," *Comput. Netw.*, vol. 194, 2021, Art. no. 108149.

[11] A. Dunmore, J. Jang-Jaccard, F. Sabrina, and J. Kwak, "A comprehensive survey of generative adversarial networks (GANs) in cybersecurity intrusion detection," *IEEE Access*, vol. 11, pp. 76071–76094, 2023.

[12] K. He, D. D. Kim, and M. R. Asghar, "Adversarial machine learning for network intrusion detection systems: A comprehensive survey," *IEEE Commun. Surv. Tutor.*, vol. 25, no. 1, pp. 538–566, Firstquarter 2023.

[13] J. C. Costa, T. Roxo, H. Proença, and P. R. Inácio, "How deep learning sees the world: A survey on adversarial attacks & defenses," *IEEE Access*, vol. 12, pp. 61113–61136, 2024.

[14] H. Baniecki and P. Biecek, "Adversarial attacks and defenses in explainable artificial intelligence: A survey," *Inf. Fusion*, vol. 107, 2024, Art. no. 102303.

[15] J. Vitorino, I. Praça, and E. Maia, "SoK: Realistic adversarial attacks and defenses for intelligent network intrusion detection," *Comput. Sci*, vol. 134, 2023, Art. no. 103433.

[16] S. Chung, S. Moon, J. Kim, J. Kim, S. Lim, and S. Chi, "Comparing natural language processing (NLP) applications in construction and computer science using preferred reporting items for systematic reviews (PRISMA)," *Autom. Constr.*, vol. 154, 2023, Art. no. 105020.

[17] O. H. Abdulganiyu, T. Ait Tchakoucht, and Y. K. Saheed, "A systematic literature review for network intrusion detection system (IDS)," *Int. J. Inf. Secur.*, vol. 22, no. 5, pp. 1125–1162, Oct. 2023.

[18] H. Satilmiş, S. Akleylek, and Z. Y. Tok, "A systematic literature review on Host-based intrusion detection systems," *IEEE Access*, vol. 12, pp. 27237–27266, 2024.

[19] M. Macas, C. Wu, and W. Fuertes, "Adversarial examples: A survey of attacks and defenses in deep learning-enabled cybersecurity systems," *Expert Syst. Appl*, vol. 238, p. 122223, 2023.

[20] M. Sabuhi, M. Zhou, C.-P. Bezemer, and P. Musilek, "Applications of generative adversarial networks in anomaly detection: A systematic literature review," *IEEE Access*, vol. 9, pp. 161003–161029, 2021.

[21] M. Mohebbi Moghaddam et al., "Games of GANs: Game-theoretical models for generative adversarial networks," *Artif. Intell. Rev.*, vol. 56, no. 9, pp. 9771–9807, 2023.

[22] C. Zhang, S. Yu, Z. Tian, and J. J. Yu, "Generative adversarial networks: A survey on attack and defense perspective," *ACM Comput. Surv.*, vol. 56, no. 4, pp. 1–35, 2023.

[23] N. Dalvi, P. Domingos, Mausam, S. Sanghai, and D. Verma, "Adversarial classification," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2004, pp. 99–108.

[24] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[25] S. Calzavara, C. Lucchese, and G. Tolomei, "Adversarial training of gradient-boosted decision trees," in *Proc. ACM Int. Conf. Inform. Knowl. Manage.*, 2019, pp. 2429–2432.

[26] S. Zhou, K. Li, and G. Min, "Adversarial example generation via genetic algorithm: A preliminary result," in *Proc. ACM Genetic Evol. Computation Conf. Companion*, 2022, pp. 469–470.

[27] H. Mun, S. Seo, B. Son, and J. Yun, "Black-box audio adversarial attack using particle swarm optimization," *IEEE Access*, vol. 10, pp. 23532–23544, 2022.

[28] L. Schott et al., "Robust deep reinforcement learning through adversarial attacks training: A survey," 2024, *arXiv:2403.00420*.

[29] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in *Proc. Pacific-Asia Conf. Knowl. Discovery Data Mining*, 2022, pp. 79–91.

[30] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. PMLR Int. Conf. Mach. Learn.*, vol. 70, 2017, pp. 214–223.

[31] X. Mao, Q. Li, H. Xie, R. Y. Lau, Z. Wang, and S. P. Smolley, Least squares generative adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017.

[32] S. C. A. Radford and L. Metz, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2015, *arXiv:1511.06434*.

[33] M. Mirza, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.

[34] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in *Proc. Adv. Neural Inform. Process. Syst. (NeurIPS)*, vol. 32, 2019.

[35] Z. Zhang, S. Liu, M. Li, M. Zhou, and E. Chen, "Bidirectional generative adversarial networks for neural machine translation," in *Proc. ACL Comput. Natural Lang. Learn.*, 2018, pp. 190–199.

[36] D. Shu, N. O. Leslie, C. A. Kamhoua, and C. S. Tucker, "Generative adversarial attacks against intrusion detection systems using active learning," in *Proc. ACM Workshop Wireless Secur. Mach. Learn.*, 2020, pp. 1–6.

[37] U. Sabeel, S. S. Heydari, K. El-Khatib, and K. Elgazzar, "Analyzing the quality of synthetic adversarial cyberattacks," in *Proc. IEEE Int. Conf. Netw. Service Manage. (CNSM)*, 2023, pp. 1–5.

[38] M. Wang, N. Yang, N. J. Forcade-Perkins, and N. Weng, "ProGen: Projection-based adversarial attack generation against network intrusion detection," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 5476–5491, 2024.

[39] R. Zhang, S. Luo, L. Pan, J. Hao, and J. Zhang, "Generating adversarial examples via enhancing latent spatial features of benign traffic and preserving malicious functions," *Neurocomputing*, vol. 490, no. C, pp. 413–430, Jun. 2022.

[40] G. Liu, W. Zhang, X. Li, K. Fan, and S. Yu, "VulnerGAN: A backdoor attack through vulnerability amplification against machine learning-based network intrusion detection systems," *Sci. China Inf. Sci.*, vol. 65, no. 7, 2022, Art. no. 170303.

[41] E. Seo, J. Kim, W. Lee, and J. Seok, "Adversarial attack of ML-based intrusion detection system on in-vehicle system using GAN," in *Proc. IEEE Int. Conf. Ubiquitous Future Netw. (ICUFN),* 2023, pp. 700–703.

[42] D. Sarkar, P. Vinod, and S. Y. Yerima, "Detection of Tor traffic using deep learning," in *Proc. IEEE Int. Conf. Comput. Syst. Appl. (AICCSA),* 2020, pp. 1–8.

[43] D. He, J. Dai, X. Liu, S. Zhu, S. Chan, and M. Guizani, "Adversarial attacks for intrusion detection based on bus traffic," *IEEE Netw.*, vol. 36, no. 4, pp. 203–209, Jul./Aug. 2022.

[44] J. Chen, D. Wu, Y. Zhao, N. Sharma, M. Blumenstein, and S. Yu, "Fooling intrusion detection systems using adversarially autoencoder," *Digit. Commun. Netw.*, vol. 7, no. 3, pp. 453–460, 2021.

[45] I. H. Salman, O. Nad Elhajj, A. Kayssi, and A. Chehab, "Mutated traffic detection and recovery: An adversarial generative deep learning approach," *Ann. Telecommun.*, vol. 77, no. 5, pp. 395–406, 2022.

[46] H. Chaitou, T. Robert, J. Leneutre, and L. Pautet, "Intrinsic weaknesses of IDSs to malicious adversarial attacks and their mitigation," in *Proc. Springer Int. Conf. Smart Bus. Technol.*, 2022, pp. 122–155.

[47] A. Jyothish, A. Mathew, and P. Vinod, "Effectiveness of machine learning based android malware detectors against adversarial attacks," *Cluster Comput.*, vol. 27, no. 3, pp. 2549–2569, 2024.

[48] M. P. Novaes, L. F. Carvalho, J. Lloret, and M. L. Proença, "Adversarial deep learning approach detection and defense against DDoS attacks in SDN environments," *Future Gener. Comput. Syst*, vol. 125, pp. 156–167, Jun. 2021.

[49] S. Zhang, X. Xie, and Y. Xu, "A brute-force black-box method to attack machine learning-based systems in cybersecurity," *IEEE Access*, vol. 8, pp. 128250–128263, 2020.

[50] R. Chauhan and S. S. Heydari, "Polymorphic adversarial DDoS attack on IDS using GAN," in *Proc. IEEE Int. Symp. Netw., Comput. Commun. (ISNCC),* 2020, pp. 1–6.

[51] M. Abdelaty, S. Scott-Hayward, R. Doriguzzi-Corin, and D. Siracusa, "GADoT: GAN-based adversarial training for robust DDoS attack detection," in *Proc. IEEE Commun. Netw. Secur. (CNS),* 2021, pp. 119–127.

[52] P. T. Duy, L. K. Tien, N. H. Khoa, D. T. T. Hien, A. G.-T. Nguyen, and V.-H. Pham, "DIGFuPAS: Deceive IDS with GAN and function-preserving on adversarial samples in SDN-enabled networks," *Comput. Sci.*, vol. 109, 2021, Art. no. 102367.

[53] Y. Zhu, L. Cui, Z. Ding, L. Li, Y. Liu, and Z. Hao, "Black box attack and network intrusion detection using machine learning for malicious traffic," *Comput. Sci.*, vol. 123, 2022, Art. no. 102922.

[54] P. T. Duy, N. H. Khoa, D. T. T. Hien, H. D. Hoang, and V.-H. Pham, "Investigating on the robustness of flow-based intrusion detection system against adversarial samples using generative adversarial networks," *J. Inf. Secur. Appl.*, vol. 74, 2023, Art. no. 103472.

[55] A. Sarıkaya, B. G. Kılıç, and M. Demirci, "RAIDS: Robust autoencoder-based intrusion detection system model against adversarial attacks," *Comput. Sci.*, vol. 135, 2023, Art. no. 103483.

[56] S. Li, J. Wang, Y. Wang, G. Zhou, and Y. Zhao, "EIFDAA: Evaluation of an IDS with function-discarding adversarial attacks in the IIoT," *Heliyon*, vol. 9, no. 2, 2023, Art. no. e13520.

[57] A. Mustapha et al., "Detecting DDoS attacks using adversarial neural network," *Comput. Sci.*, vol. 127, 2023, Art. no. 103117.

[58] H. D. Hoang, D. T. Thu Hien, T. B. Xuan, T. N. Ngoc Minh, P. T. Duy, and V.-H. Pham, "DA-GAN: Domain adaptation for generative adversarial networks-assisted cyber threat detection," in *Proc. IEEE Int. Conf. Comput. Commun. Technol. (RIVF)*, 2022, pp. 29–34.

[59] Y. Zhong, Y. Zhu, Z. Wang, X. Yin, X. Shi, and K. Li, "An adversarial learning model for intrusion detection in real complex network environments," *Proc. Int. Conf. Wireless Algorithms, Syst., Appl. (WASA)*, 2020, pp. 794–806.

[60] R. Yumlembam, B. Issac, S. M. Jacob, and L. Yang, "Comprehensive botnet detection by mitigating adversarial attacks, navigating the subtleties of perturbation distances and fortifying predictions with conformal layers," *Inf. Fusion*, vol. 111, 2024, Art. no. 102529.

[61] A. Grini, B. E. Khamlichi, A. E. Afia, and A. E. Fallah-Seghrouchni, "HPAC-IDS: A hierarchical packet attention convolution for intrusion detection system," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2024, pp. 1–6.

[62] S. Laudanna, A. Di Sorbo, P. Vinod, C. A. Visaggio, and G. Canfora, "Transformer or autoencoder? who is the ultimate adversary for attack detectors?," *Int. J. Inf. Secur.*, vol. 24, no. 1, p. 26, 2025.

[63] J. Wang, X. Chang, Y. Wang, R. Rodrıguez, and J. Zhang, "LSGAN-AT: Enhancing malware detector robustness against adversarial examples," *Cybersecurity*, vol. 4, pp. 2523–3246, Dec. 2021.

[64] F. Chen, Y. Shang, J. Hu, and B. Xu, "Few features attack to fool machine learning models through mask-based GAN," in *Proc. IEEE Int. Joint Conf. Neural Netw. (IJCNN)*, 2020, pp. 1–8.

[65] M. Bai et al., "Adversarial attack against intrusion detectors in cyber-physical systems with minimal perturbations," in *Proc. IEEE Int. Symp. Parallel Distrib. Process. Appl. (ISPA)*, 2024, pp. 816–825.

[66] Z. Liu, Q. Ye, and F. Cui, "Sophon IDS: Mitigating the effectiveness of GAN-based adversarial attacks via tailored misinformation," in *Proc. Wireless Artif. Intell. Comput. Syst. Appl.*, vol. 14997, 2025, pp. 415–427.

[67] Z. He, Y. Chen, A. Chen, D. Zhang, H. Zhang, and J. Zhang, "GAN-enabled U-shaped network for adversarial attack generation for autonomous unmanned vehicles," *IEEE Trans. Autom. Sci. Eng.*, vol. 23, pp. 3399–3409, 2026.

[68] P. Dadkhah, S. Ahmadi, and M. R. Aref, "GenAtt NIDS attack: A practical generative adversarial attack against network intrusion detection systems," in *Proc. IEEE Inte. Comp. Conf., Comp. Soc. Iran (CSICC)*, 2025, pp. 1–7.

[69] M. Pujari, B. P. Cherukuri, A. Y. Javaid, and W. Sun, "An approach to improve the robustness of machine learning based intrusion detection system models against the Carlini-Wagner attack," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience (CSR)*, 2022, pp. 62–67.

[70] T. Saravanan, S. Deepa, and P. Sasikumar, "Advanced EGAN-IDS framework for resilience against adversarial attacks using Multi-headed attention module," *Proc. Comput. Sci*, vol. 230, pp. 203–213, Jan. 2023.

[71] D. C. Asimopoulos, et al., "Breaching the defense: Investigating FGSM and CTGan adversarial attacks on IEC 60870-5-104 AI-enabled intrusion detection systems," in *Proc. ACM Int. Conf. Availability, Rel. Secur.*, 2023, pp. 1–8.

[72] Y. Zhang, Y. Zhang, Z. Zhang, H. Bai, T. Zhong, and M. Song, "Evaluation of data poisoning attacks on federated learning-based network intrusion detection system," in *Proc. IEEE Int. Conf. High Perform. Comput. & Commun.; Int. Conf. Data Sci. & Systems; Int. Conf. on Smart City; Int. Conf. on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys)* 2022, pp. 964–971.

[73] Y. Alslman, M. Alkasassbeh, and M. Almseidin, "A robust SNMP-MIB intrusion detection system against adversarial attacks," *Arab. J. Sci. Eng.*, vol. 49, no. 3, pp. 4179–4195, 2024.

[74] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.

[75] A. Paya, S. Arroni, V. García-Díaz, and A. Gómez, "Apollon: A robust defense system against adversarial machine learning attacks in intrusion detection systems," *Comput. Secur.*, vol. 136, 2024, Art. no. 103546.

[76] H. Benaddi, M. Jouhari, K. Ibrahimi, A. Benslimane, and E. M. Amhoud, "Adversarial attacks against IoT networks using conditional GAN based learning," in *Proc. IEEE Global Commun. (GLOBECOM)*, 2022, pp. 2788–2793.

[77] N. Singh and T. Sandhan, "Learnable GAN regularization for improving training stability in limited data paradigm," in *Proc. Springer Nature Switzerland Comput. Vis. Image Process.*, 2024, pp. 542–554.

[78] A. AlEroud and G. Karabatis, "SDN-GAN: Generative adversarial deep NNs for synthesizing cyber attacks on software defined networks," in *Proc. Springer OTM Confederated Int. Workshops Move to Meaningful Internet Syst.*, 2020, pp. 162–174.

[79] F. Cui, Q. Ye, and P. Kibenge-MacLeod, "A Wasserstein GAN-based framework for adversarial attacks against intrusion detection systems," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2023, pp. 3187–3192.

[80] J. Wang, Q. Liu, C. Liu, and J. Yin, "GAN-based adversarial patch for malware C2 traffic to bypass DL detector," in *Proc. Springer Int. Conf. Inform. Commun. Secur.*, 2021, pp. 78–96.

[81] E. Alhajjar, P. Maxwell, and N. Bastian, "Adversarial machine learning in network intrusion detection systems," *Expert Syst. Appl.*, vol. 186, 2021, Art. no. 115782.

[82] J. Chen, X. Gao, R. Deng, Y. He, C. Fang, and P. Cheng, "Generating adversarial examples against machine learning-based intrusion detector in industrial control systems," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 3, pp. 1810–1825, May/Jun. 2020.

[83] C. Reilly, S. O Shaughnessy, and C. Thorpe, “Robustness of image-based malware classification models trained with generative adversarial networks,” in *Proc. ACM Eur. Interdisciplinary CyberSecur. Conf.*, 2023, pp. 92–99.
[84] A. S. Dina, A. B. Siddique, and D. Manivannan, “Effect of balancing data using synthetic data on the performance of machine learning classifiers for intrusion detection in computer networks,” *IEEE Access*, vol. 10, pp. 96731–96747, 2022.
[85] R. Mohammad, F. Saeed, A. A. Almazroi, F. S. Alsubaei, and A. A. Almazroi, “Enhancing intrusion detection systems using a deep learning and data augmentation approach,” *Systems*, vol. 12, no. 3, p. 79, 2024.
[86] A. Islam, S.-Y. Chang, J. Kim, and J. Kim, “Anomaly detection in 5G using variational autoencoders,” in *Proc. IEEE Silicon Valley CyberSecur. Conf. (SVCC)*, 2024, pp. 1–6.
[87] X. Yang and L. Wang, “SDP-CycleGAN: CycleGAN based on siamese data pairs for unraveling data imbalance problems in anomaly detection,” in *Proc. IEEE Int. Conf. Adv. Cloud Big Data (CBD)*, 2022, pp. 151–157.
[88] M. Asif, M. AshiqUR Rahman, K. Akkaya, H. Shahriar, and A. Cuzzocrea, “Adversarial data-augmented resilient intrusion detection system for unmanned aerial vehicles,” in *Proc. IEEE Int. Conf. Big Data (BigData)*, 2023, pp. 5428–5437.
[89] M. Adiban, S. M. Siniscalchi, and G. Salvi, “A step-by-step training method for multi generator GANs with application to anomaly detection and cybersecurity,” *Neurocomputing*, vol. 537, pp. 296–308, Jun. 2023.
[90] D. Li, D. Chen, B. Jin, L. Shi, J. Goh, and S.-K. Ng, “MAD-GAN: Multivariate anomaly detection for time series data with generative adversarial networks,” in *Proc. Artif. Neural Netw. Mach. Learn. (ICANN)*, 2019, pp. 703–716.
[91] H. Zhang, S. Xu, J. Jiao, P. Xie, R. Salakhutdinov, and E. P. Xing, “Stackelberg GAN: Towards provable minimax equilibrium via multi-generator architectures,” 2018, *arXiv:181108010*.
[92] M. Brückner and T. Scheffer, “Stackelberg games for adversarial prediction problems,” in *Proc. ACM Int. Conf. Knowl. Discovery Data Mining (SIGKDD)*, 2011, pp. 547–555.
[93] K. N. Kumar, C. K. Mohan, and L. R. Cenkeramaddi, “The impact of adversarial attacks on federated learning: A survey,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 2672–2691, May 2024.
[94] S. Samarakoon et al., “5G-NIDD: A comprehensive network intrusion detection dataset generated over 5G wireless network,” 2022, *arXiv:2212.01298*.
[95] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, “CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment,” *Sensors*, vol. 23, no. 13, Jun. 2023.

Yasmeen Alslman received the B.Sc. degree in computer science from the University of Jordan, Amman, Jordan, in 2018. She received the M.Sc. degree in computer science in 2021 from Princess Sumaya University for Technology (PSUT), Amman, where currently she is working toward the Ph.D. degree with the Computer Science Department.

Her research interests include the broad areas of cybersecurity and artificial intelligence, with a particular focus on adversarial machine learning, intrusion detection systems, and cryptography.

Mouhammd Alkasassbeh received the Ph.D. degree in computer science from the School of Computing, Portsmouth University, Portsmouth, England, U.K., in 2008.

He holds a full professorship with the Computer Science Department, Princess Sumaya University for Technology, Amman, Jordan. His research interests include network traffic analysis, network fault detection, network fault and anomaly classification, as well as the application of machine learning within the realm of computer networking and network security.

Mohammad J. Abdel-Rahman (Senior Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Arizona, Tucson, AZ, USA.

He is currently a Professor with the Data Science Department, Princess Sumaya University for Technology, Amman, Jordan, and an Adjunct Assistant Professor with the ECE Department, Virginia Tech, Blacksburg, VA, USA. His research interests include the integration of artificial intelligence and operations research to tackle complex problems in fields such as networking, communications, energy, healthcare, and transportation.

Prof. Abdel-Rahman is a member of the Global Young Academy. He currently serves as an Associate Editor for IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING (AI-empowered resource management track), IEEE ACCESS, and Wireless Personal Communications (Springer Nature). He also regularly contributes to the technical program committees of major international conferences.